Data sheet

SOMATOM X.cite

syngo CT VB10

siemens-healthineers.com/somatom-xcite

International version.
Not for distribution or use in the U.S.

SIEMENS Healthineers

Intelligent imaging. Excellence empowered.

As the number and complexity of radiological procedures increases, demands on staff are reaching unsustainable levels. This continues to impact consistent image quality. Although our advanced CT systems have the potential to expand precision medicine, too often this potential remains untapped.

SOMATOM® X.cite changes that. Together with myExam Companion, it launches the era of intelligent imaging. Now users of any skill level can unlock their system's groundbreaking potential. myExam Companion uses the new possibilities of digitalization to turn data into built-in expertise. This transforms real-time patient characterization – and the way staff operate their CT scanner. myExam Companion guides them through any procedure, supporting with the optimization of parameters for individual patients. For technologists, it makes interacting with patients and the system easier and more natural. For radiologists, it reduces unwarranted variations and generates consistent, comprehensive results.

SOMATOM X.cite has a patient-friendly, 82 cm bore and a Vectron® X-ray tube – for unprecedented power and resolution. This makes it exceptionally equipped to improve personalization, no matter how challenging your patient or throughput. From routine exams to advanced quantitative and functional procedures, SOMATOM X.cite empowers excellence in radiology.

Highlights

Intelligent navigation for enhanced consistency

No matter the user, patient, or throughput, our Companions guide users through even time-critical procedures, including emergencies and interventions. This helps them interact easily, naturally, and precisely with both patient and technology.

- The automation of myExam Companion helps technologists individualize parameters and radiologists analyze images efficiently even in time-critical situations like triage of emergency cases.
- For interventions, myNeedle Companion delivers laser-assisted multiple-needle path planning, guided insertion, and near-full in-room control.

Personalized imaging for consistent, comprehensive results

SOMATOM X.cite has the power and user guidance to generate the comprehensive information that can help radiologists diagnose with precision and confidence.

The power stems from its outstanding imaging chain, which includes the Vectron X-ray tube. The user guidance comes from myExam Companion, which tailors acquisition to the individual patient. It helps technologists generate high-quality results in accordance with the latest guidelines, quickly and consistently. It even calculates calcium scores automatically, with zero additional clicks.

Patient-friendly design with an 82 cm bore

More natural communication with patients and our Mobile Workflow help patients relax – in combination with the large bore and enhanced accessibility of the new multi-purpose table.

Consistent standards across your institution

myExam Companion, myNeedle Companion, and Shui® – the design system of Siemens Healthineers – provide users with common interfaces across multiple modalities. Combined with our digital solutions, they can help you redefine and standardize protocols across your institution, optimizing clinical workflow, staffing schedules, results, and productivity.

System configuration

Standard HW configuration

- Vectron X-ray tube
- 82 cm bore size
- Stellar^{Infinity} detector
- Diagonal z-Sharp technology
- Adaptive Dose Shield
- Tin Filter
- 0.3[1], 0.5, 1.0 s rotation time
- Tablet for mobile workflow
- Remote control

Generator
- 105 kW
- High Power (1,200 mA) @ 70 & 80 kV
- 10 kV Steps (70–150 kV)

Acquisition Workplace
- 24" / 60 cm flat-screen monitor
- External USB 3.0 disk support

HW configurations

Patient Table Options
- 227 kg / 500 lbs table load; 160 cm scannable range
- 307 kg / 676 lbs table load; 200 cm scannable range
- 340 kg / 750 lbs table load; 200 cm scannable range

Cooling Options
- Air cooling
- Water cooling

Computers – Standard Configuration
- ICS X. Standard
- IRS X. Standard

Computers – Power Configuration
- ICS X. Power
- IRS X. Power

Additional optional HW

FAST Integrated Workflow
FAST 3D Camera

Mobile workflow support
- Rear panel with control panels and charging docks for tablets
- Extra tablets (up to 4 supported in total)
- Wall / table docking stations as additional charging spots for the tablets
- Integrated Injector Arm

Patient Experience Options
- Ring Moodlight
- Funnel Moodlight
- Patient Observation Camera
- Visual Patient Instruction

Acquisition Workplace with Dual Monitor
Dual 24" / 60 cm flat-screen monitor with dual display functionality

Additional Table Options
- Patient table foot switch
- X-ray foot switch
- Patient table 2000 mm / 307 kg RT
- Multi-purpose Table RT
- 307 kg Multi-index RTP Overlay
- Trauma Overlay

myExam Satellite
Additional workplace sharing database and applications with the main Acquisition Workplace

[1] *Optional*

System configuration

Standard SW and application packages

- *syngo* Examination
- *syngo* Archiving & Network
- SureView
- Dynamic Serio Scan
- HD FOV
- Video Capture and Editing Tool
- Screen recorder
- Exam Designer
- WorkStream4D
- Adaptive Signal Boost

myExam Companion
- myExam Compass
- myExam Cockpit

GO Technologies
- Scan&GO tablet application
- Check&GO:
  - Coverage
  - Contrast media
  - Metal detection
- Recon&GO – Inline Results, including:
  - Anatomical ranges (Parallel / Radial)
  - Table removal
  - Bone removal
  - Vessel Ranges (aorta, run-offs, carotid arteries)
  - Spine Ranges
  - Rib Ranges
  - Multi Recon
- CT View&GO, including:
  - 2D and 3D (MPR, MIP, VRT)
  - Evaluation tools
  - Filming
  - Vessel Extension
  - Endoscopic View
  - Lung Nodules Segmentation
  - Diameter / WHO area
  - ROI HU Threshold
  - Spine Ranges
  - Average

FAST Technologies
- FAST Planning
- FAST Adjust
- FAST ROI
- FAST Contact

CARE Technologies
- CARE kV
- 10 kV Steps
- CARE Child
- CARE Dose4D
- CARE Topo
- CARE Profile
- CARE Filter
- CARE Bolus CT
- X-CARE
- ADMIRE®[1]
  (Advanced Modeled Iterative Reconstruction)
- Flex Dose Profile
- Protocol Password Protection
- DICOM SR Dose Reports
- DoseLogs
- Dose Notification
- Dose Alert

IT Security
syngo System Security

[1] *Image quality as defined by low-contrast detectability using a model observer method for evaluation. As demonstrated using SOMATOM Force data, equivalent low-contrast detectability can be achieved with 80% to 85% less dose using ADMIRE at the highest strength level for thin (0.6 mm) reconstruction slices in measured and simulated body and head phantoms for low-contrast objects with different contrasts. In clinical practice, the use of ADMIRE may reduce CT patient dose depending on the clinical task, patient size, anatomical location, and clinical practice. A consultation with a radiologist and a physicist should be made to determine the appropriate dose to obtain diagnostic image quality for the particular clinical task. See ADMIRE data sheet for further information.*

System configuration

Optional SW and application packages

iMAR
iterative Metal Artifact Reduction

CARE Application
CARE Contrast III

Cardiac Imaging
- Physiological Measurement Module
- ECG cable
- Cardio Spiral scan mode
- Adaptive Cardio Sequence scan mode
- Cardio BestPhase
- ZeeFree
- Recon&GO – Inline Results, including:
  - CaScoring
  - Cardiac Ranges
  - Vessel Ranges (LAD, RCA, CX)
  - Heart Isolation
  - Coronary Tree
- CT View&GO, including:
  - Heart Isolation
  - Coronary Tree
- *syngo*.CT CaScoring application for AWP

Spectral Imaging with Dual Energy
- TwinSpiral Dual Energy
- TwinBeam Dual Energy
- Recon&GO – Inline Results, including:
  - DE Ranges (Parallel / Radial)
  - DE SPP (Spectral Post-processing)
- Recon&GO – Spectral Recon
- CT View&GO, including:
  - DE ROI
  - Interactive Spectral Imaging
- *syngo*.CT Dual Energy applications for AWP

CT LungCAD
- Recon&GO – Inline Results, including:
  - LungCAD
- CT View&GO, including
  - LungCAD

Precision Matrix
- 1024 x 1024 and 768 x 768 reconstructions

Neuro Imaging
- Flex 4D Spiral – Neuro[1]
- Recon&GO – Inline Results, including:
  - ASPECTS
- CT View&GO, including:
  - Neuro DSA
  - Stroke Layouts
- Recon&GO – Inline Neuro Perfusion
- *syngo*.CT Neuro Perfusion application for AWP

Advanced 4D Imaging[1,2]
Flex 4D Spiral – Body

Emergency Imaging
- Trauma layouts
- Recon&GO – Inline Skull Unfolding
- Recon&GO – Inline Brain Hemorrhage
- Recon&GO – Inline Brain Hemorrhage SAH

myNeedle Companion for CT-guided interventions
- myNeedle Guide 3D incl. myNeedle Detection and myNeedle Guide 2D with i-Spiral and FAST i-Sequence scan modes
- myNeedle Laser
- i-Fluoro
- HandCARE
- i-Joystick
- X-Ray Foot switch
- 24" or 32" in-room monitor either mounted at the ceiling or on a cart
- Tablet dock for patient table
- Table side rails long

CT Dental
syngo.CT Dental application for AWP

CT Osteo
- CT View&GO, including:
  - CT Osteo
- Mattress phantom

Radiation Oncology
- Direct i4D[3]
- Radiation Therapy Basic
- Respiratory Motion Management
- FAST 4D
- Varian RGSC interface
- ANZAI interface
- Open interface
- DirectDensity
- DirectORGANS
- DirectORGANS Advanced
- RadOnc Excellence
- Multimodality Deformable Registration
- Direct Laser Steering

Fleet management
- *syngo* Expert-i
- *syngo* Virtual Cockpit

[1] *Not compatible with 227 kg table*
[2] *Recommended applications for evaluation in syngo.via: syngo.CT Body Perfusion and syngo.CT Dynamic Angio*
[3] *Requires Varian RGSC Interface or ANZAI Interface and Respiratory Motion Management option*

System hardware

Gantry

Aperture
82 cm / 32"

Depth
< 100 cm / 39"

Distance scan plane to gantry front cover
- 353 mm / 14"
- The short distance from the gantry front to the scan plane allows for easy operator access.

Distance focal spot to isocenter
61 cm / 24"

Distance focal spot to detector
111.3 cm / 43.82"

Scan field
- 50 cm / 19.7"
- 81.5 cm / 32.09" with HD FOV[1]

Physical tilt
- Up to ± 25°
- Up to ± 30° with multi-purpose table[2]

Rotation time
0.3[2], 0.5, 1.0 s

Temporal resolution
- Down to 150[2] ms native temporal resolution
- Down to 75[2] ms bi-segmented temporal resolution

Gantry front control panels
2 sets of control panels at the front and 2 at the back[2] for convenient and fast patient positioning

Three laser light markers
Coronal, sagittal, axial laser light showing the isocenter position of the scan plane.

Integrated injector arm[2]
The gantry-mounted injector arm lets you position the injector where you need it, when you need it. Whereas a traditional injector cart is often in the way, the injector arm makes for a neat and organized working environment and still lets you flexibly arrange the injector.

[1] *The image quality for the area outside the 50 cm scan field of view does not meet the image quality of the area inside the 50 cm scan field of view. Image artifacts may appear, depending on the patient setup and anatomy scanned. HD FOV cannot be used to scan FOV smaller than 50 cm.*
[2] *Optional*
[3] *First tablet standard, additional tablets optional*

Mobile workflow support
- 2 tablet cradles at the front and 2 at the back[2] for magnetic docking and charging of 4 tablets simultaneously
- Up to 5 tablets[3]
- Remote control for more flexible operation
- Wall-mounted / desk docking stations as additional charging spots for the tablets[2] and remote control

Patient experience options

Patient Observation Camera
The camera is directly integrated in the gantry funnel, in order to offer a bore view for a closer look at the patient during the whole examination. This is an important patient benefit, because the patient can be observed when they are inside the gantry – when it matters the most. This potentially avoids the need to install a monitoring camera in the scan room, reducing the initial investment. At the same time, this improves the technologists' confidence for patient-centric care and for developing patient trust.

Visual Patient Instruction (VPI)
Easy-to-follow visual instructions and an intuitive graphical breathhold count-down displayed on the front and rear part of the tunnel to help patients comply with breathhold times.

Moodlight
Adapt the color environment for the patient with the lighting at the gantry ring.
Light up the scanner funnel with different colors to enhance well-being by creating the impression of a bigger space.

System hardware

Tube assembly

Tube
The Vectron X-ray tube provides high power reserves at every kV value (up to 1,200 mA for 70 and 80 kV) allowing you to routinely perform low kV exams even in adults. This may potentially reduce the amount of contrast media needed, thus enhancing patient safety.

Tube current range
10–1,200 mA

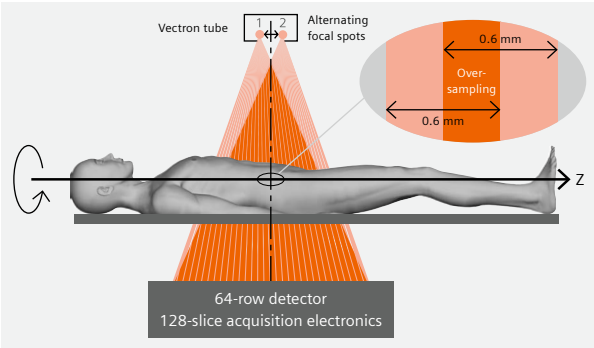
Tube voltage
- 70, 80, 90, 100, 110, 120, 130, 140, 150 kV
- Automatically selected through CARE kV based on patient size and clinical task

Voltage [kV] with maximal available tube current [mA]
70 at 1,200, 80 at 1,200, 90 at 1,166, 100 at 1,050, 110 at 954, 120 at 875, 130 at 807, 140 at 750, 150 at 700

Tube anode heat storage capacity
Higher than 30 MHU[1] (equivalent value compared to the performance of a conventional tube)

Focal spot size according to IEC 60336
- 0.6 x 0.7 / 8°
- 0.8 x 1.1 / 8°

z-Sharp technology



The unique Vectron X-ray tube utilizes an electron beam that is accurately and rapidly deflected, creating two precise focal spots alternating 4,032 times per second. This doubles the X-ray projections reaching each detector element. The two overlapping projections result in an oversampling in the z-direction and allow acquiring twice the number of slices per detector row.

The resulting measurements interleave half a detector slice width, doubling the scan information without a corresponding increase in dose. This provides a scan-speed-independent spatial resolution in the z-direction of up to 0.30 mm and a corresponding reduction of spiral artifacts in the daily clinical routine at any position within the scan field.

Temperature monitoring
Computer-controlled monitoring of anode Temperature

Tube cooling rate
2.7 MHU / min

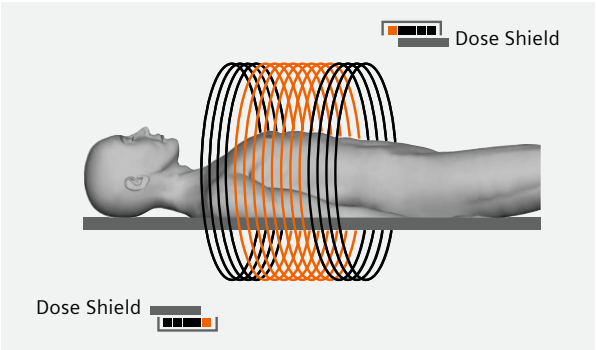
[1] *The Vectron X-ray tube exceeds the performance of a conventional 30 MHU tube*

System hardware

Generator

Max. Power
105 kW

Adaptive Dose Shield



- Dynamic tube collimation that protects the patient from clinically irrelevant radiation in all standard spiral CT scan modes.
- Avoid unnecessary radiation most effectively in short scan ranges, especially in cardiac and pediatric scans.

Tin Filter

Inherited from high-end dual-source scanners, Tin Filter technology cuts out lower energies to reduce dose and optimizes contrast between soft tissue and air.

This has direct benefits for imaging areas such as the lungs, colon, and sinuses. In addition, clinical experience shows that Tin Filter technology reduces beam-hardening artifacts and improves image quality in bony structures, which means it is also extremely useful in orthopedic examinations. As a result, you get CT imaging at exceptionally low dose levels, comparable to conventional X-ray.

Tin Filter technology protects you and your patients with ultra-low doses during intervention. Factory protocols for low-dose lung cancer screening, colon and sinus and calcium scoring employing the tin filter. Provide dose savings, even during the topogram. Only Siemens Healthineers CT scanners enable lung imaging powered by Tin Filter technology.

System hardware

Data Acquisition System

Stellar^{Infinity} Detector

SOMATOM X.cite features a Stellar^{Infinity} detector boosting an Integrated Circuit Detector technology, where photodiode and electronics are integrated on one single integrated circuit. It has been shown in literature that this integrated design allows superior imaging compared to conventional detector circuit designs, supporting, e.g.:

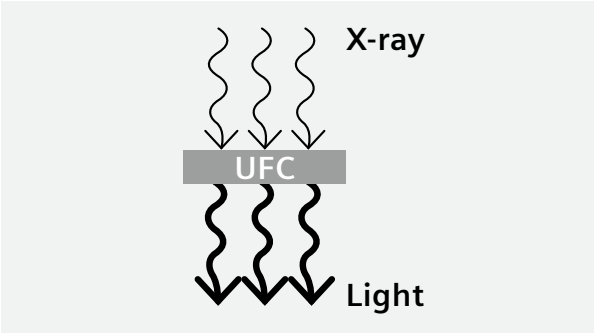
- Superior objective and subjective image quality in head CTs
- Reduced image noise and streak artifacts, especially in low dose or low kV imaging or in high attenuation areas such as the shoulder and pelvis regions
- Improved image quality and low-contrast detectability in abdominal CT of overweight or obese patients
- Lower image noise and improved image quality in coronary CTA and coronary stent imaging

Due to the Integrated Detector Circuit design of the Stellar^{Infinity} detector, electronic components (microchips, conductors, etc.) are integrated directly at the photo diode. This reduces electronic noise coming from the detector elements and thus minimizes the negative impact of electronics noise to image quality. TrueSignal technology for minimized electronic noise.

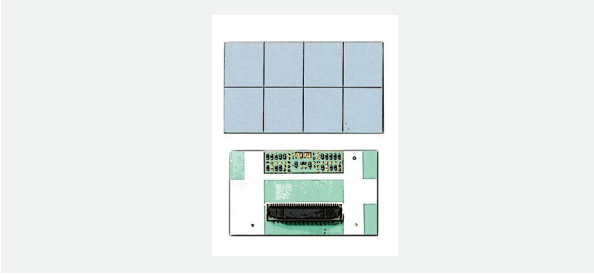
Visualize fine details thanks to new geometry with higher channel density in scan plane. The small detector apertures of the Stellar^{Infinity} detector in combination with the in-plane and z-axis flying focal spot enable excellent in-plane and through-plane sampling, achieving high image resolution with a new geometry and 840 channels in the scan plane. This equates to a pixel length of 0.47 mm and an exceptional degree of precision.

Stellar^{Infinity} is integrated with ADMIRE and allows to reconstruct 0.5 mm slices with 0.3 mm cross-plane resolution.

UFC (Ultra Fast Ceramics)



Edge technology



HiDynamics

HiDynamics allows an extended dynamic range.

Due to the full electronic integration dynamic bandwidth is significantly extended, allowing the detector signal to be transmitted in greater precision.

On a spherical detector, the radiation sensitive area has a spherical shape instead of the cylindrical shape, common in traditional CTs. In the SOMATOM X.cite Stellar^{Infinity} detector, this spherical detector shape is combined with a 2D anti-scatter grid. This innovation is called a 3D collimator grid.

System hardware

Adaptive Signal Boost
The Adaptive Signal Boost amplifies low signal areas of the CT data when high attenuation is present – such as when imaging obese patients or patients with metal implants or pediatric imaging at low kV.

Max. number of slices / rotation
- 128 (acquired slices through diagonal z-Sharp technology)
- max. 384 (reconstructed slices)

Number of detector rows
64

Number of detector elements
53,760

Number of detector channels per row (in-plane)
840 elements (1,680 channels per row)

Number of projections 1 s / 360°
Up to 4,032

Sequence acquisition modes
128 x 0.6 mm, Sn128 x 0.6 mm,
64 x 0.6 mm, Sn64 x 0.6 mm, 1 x 10 mm,
1 x 5 mm, 3 x 3 mm, Sn3 x 3 mm,
3 x 5 mm, Sn3 x 5 mm

Spiral acquisition modes
128 x 0.6 mm, Sn128 x 0.6 mm,
64 x 0.6 mm, Sn64 x 0.6 mm,
AuSn64 x 0.6 mm[1] (TwinBeam Dual Energy modes)

Patient table 160 cm / 227 kg

Max. table load
227 kg / 500 lbs

Max. table feed speed
3–250 mm/s

Vertical table travel range
53.9–96.4 cm / 21.2–38" (transfer height)

Vertical travel speed
35 mm/s

Scannable range
160 cm / 63" with patient table extension[1]

Patient table 200 cm / 307 kg[1]

Max. table load
307 kg / 676 lbs

Max. table feed speed
3–250 mm/s

Vertical table travel range
56.8–95.3 cm / 22.4–37.5" (transfer height)

Vertical travel speed
35 mm/s

Scannable range
200 cm / 78.7" with patient table extension[1]

Multi-purpose patient table 200 cm / 340 kg[1]

Max. table load
340 kg / 750 lbs

Max. table feed speed
3–262 mm/s

Vertical table travel range
43.2–101.8 cm / 17–40" (transfer height)

Vertical travel speed
50 mm/s

Scannable range
200 cm / 78.7" with patient table extension[1]

Allows gantry tilt of up to ± 30°

Table options

Patient table foot switch
- Foot switch located on the bottom edge of the patient table allowing table positioning.
- Speeds up patient preparation and keeps operator hands sterile.

307 kg Multi-index RTP overlay[2]
- Multi-indexing with Varian and Elekta indexing
- Light weight overlay
- TG-66 compliant

Table Side Rails short
Bilateral pleurical short table side rails for convenient mounting of accessories

Table Side Rails long
Bilateral pleurical long table side rails for convenient mounting of accessories,in particular to mount dedicated interventional hardware

[1] *Optional*
[2] *Requires patient table 2,000 mm / 307 kg RT*

The new workplace design

The gantry-integrated ICS (Image Control System) gives you complete flexibility over where you position the workstation. Depending on your needs and infrastructure, you can set it up in the same room, outside the scan room, or in a separate control room. The optional workplace myExam Satellite brings additional workflow flexibility without interrupting the scanning program.

Acquisition Workplace (AWP)

Computer integrated into the gantry
Hardware integrated into the gantry to:
- Enable flexible room design (see section on installation)
- Minimize the elements of the new workplace design to a monitor, keyboard, mouse and the control box

Additional storage
External USB 3.0 disks for quick and easy raw data storage are supported

ICS X. Standard

High-performance computer CPU
Intel Xeon

RAM
Min. 32 GB DDR4 RAM

Hard disk
Min. 960 GB

Image storage
Min. 800,000 (400 GB)

ICS X. Power

High-performance computer CPU
Intel Xeon 6126

RAM
128 GB DDR4 RAM

Hard disk
3,840 GB

Image storage
5,695,000 (2,800 GB)

Image reconstruction

Real-time display
- Real-time image display (512 x 512) during spiral acquisition on the workplace
- Wireless transfer of images for preview on the tablet. Transfer starts immediately after the end of scanning

Slice thickness
0.5–10 mm

Recon field[1]
- 5–50 cm / 1.9–19.7"
- 5–81.5 cm / 1.9–32.09" with HD FOV[2]

Recon matrix (fix axial)
- 512 x 512
- 768 x 768[3]
- 1024 x 1024[3]

[1] *5– 42.1 cm FOV with up to 1.7 pitch; above 42.1 cm FOV with up to 0.8 pitch*
[2] *The image quality for the area outside the 50 cm scan field of view does not meet the image quality of the area inside the 50 cm scan field of view. Image artifacts may appear, depending on the patient setup and anatomy scanned. HD FOV cannot be used for scan FOV smaller than 50 cm.*
[3] *Optional*

The new workplace design

HU scale
−8,192 to +57,343

Advanced algorithms
- Iterative Beam Hardening Correction (iBHC) for reduction of beam-hardening artifacts, e.g., in head images
- Large selection of reconstruction kernels to adapt to specific clinical needs
- Any kV CaScoring is a specific reconstruction kernel to perform Agatston equivalent scores, even at lower kV settings where dose saving potential can be significant

Wide range of freely selectable slice thicknesses for prospective and/or retrospective reconstruction

Standard monitor
- 24.1"/61.1 cm flat screen
- 1,920 x 1,200 resolution

Additional monitor[1]

Dual monitor[1]

IRS X. Standard

CPU
Intel Xeon

Memory
32 GB

Storage
>1,900 GB

Reconstruction rate
- 75 fps for FBP
- 60 fps for ADMIRE

IRS X. Power

CPU
Intel Xeon

Memory
96 GB

Storage
>5,900 GB

Reconstruction rate
- 90 fps for FBP
- 80 fps for ADMIRE

myExam Satellite[1]

Additional workplace sharing database and applications with the main Acquisition Workplace for a more flexible workflow
- While the AWP is being used for protocol preparation or scanning, myExam Satellite can be simultaneously used for filming, results creation or image interpretation
- Results and post-processing are simultaneously available at the AWP and myExam Satellite, including Recon&GO – inline Skull Unfolding[1] and Brain Hemorrhage[3]
- Same applications as available at the AWP for different clinical areas, including Dual Energy[1], Neuro Perfusion[1] and trauma layouts[1]
- Remote Recon, enabling the possibility to perform RAW data reconstruction directly at the myExam Satellite.

[1] *Optional*

Standard system software and applications

syngo Examination

Exam designer

Easy and intuitive way to change and manage scan protocols

Topogram

Length
- 128–1,680 mm / 5–63" with table extension[1]
- 128–2,080 mm / 5–78.7" with table extension[1]

Scan speed
20 cm/s

Scan times
- 1.36–8.76 s
- 1.36–10.76 s[1]

Views
a.p., p.a., lateral

Real-time topogram

Manual interruption possible once desired anatomy has been imaged

Topogram scan using Tin Filter for further dose reductions

Patient communication

Automatic Patient Instruction (API)
- Freely recordable
- 7 API types per language
- > 50 languages
- New gentle voice for better patient comfort and increased audibility

Integrated patient intercom

Sequence acquisition

Reconstructed slice widths
0.5, 0.6, 0.8, 1.0, 1.5,
2, 3, 4, 5, 6, 7, 8, 10 mm

Partial scan times (240°)
0.2[1], 0.33, 0.67

Scan times (full scan)
0.3[1], 0.5, 1.0 s

Acquisition with or without table feed

Dynamic Serio Scan

Automatic clustering of scans

Multislice spiral acquisition

Reconstructed slice widths
0.5, 0.6, 0.8, 1, 1.5,
2, 3, 4, 5, 6, 7, 8, 10 mm

Temporal resolution
- 150[1] ms
- 75[1] ms (bisegment)

Scan times, full scan (360°)
0.3[1], 0.5, 1.0 s

Reconstruction increment
Min. 0.1 mm

Pitch factor
- 0.15–1.7
- Down to 0.03
  (optional with Respiratory Motion Management)

Spiral scan time
Max. 200 s

WorkStream4D

With Workstream4D, thin slice data reconstruction is not required prior to the production of reformatted images.

This enhancement saves time when compared to alternative MPR techniques 4D workflow with direct generation of axial, sagittal, coronal, or double-oblique images from standard scanning protocols.

Elimination of manual reconstruction steps and reduction of data volume, since virtually all diagnostic information is captured in 3D slices

[1] *Optional*

Standard system software and applications

syngo Examination

Patient registration

Direct input of patient information on the workplace immediately prior to scan

Pre-registration of patients at any time prior to scan

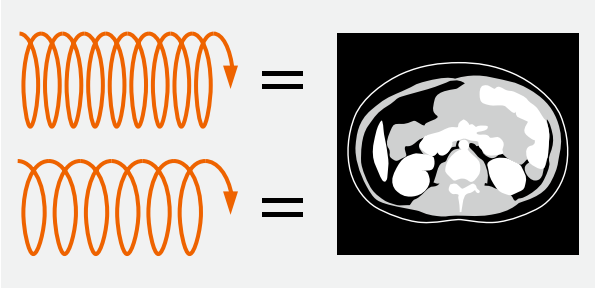
Special emergency patient registration (allows examination without entering patient data before scanning)

Transfer of patient information from HIS / RIS via DICOM Get Worklist

Transfer of examination information from scanner into HIS / RIS via MPPS (Modality Performed Procedure Step)

Automatic organization of exams according to MMPS for enhanced billing transparency

SureView: Siemens Healthineers' patented solution for multislice CT reconstruction



Pitch-independent image quality

SureView ensures that image quality is kept constant for all scan speeds, independent of the selected volume pitch.

There is higher pitch accuracy with settings available in steps of 0.1, simplifying processes by handling complex parameter settings.

Auto Field of View Adaption

When positioning the scan range, the width of the range is automatically adapted to cover the whole body of the patient.

CINE Display

Display of image sequences

Automatic or interactive with mouse control

Max. image rate: 30 frames/s

Screen recorder

Integrated solution for imaging and visualization of 4D information, allowing the generation and editing of video files for improved diagnoses, recording, and teaching. A wide range of multimedia formats are supported, e.g., AVI, Flash (SWF), GIF, QuickTime (MOV), streaming video.

Image transfer / networking

- Interface for transfer of medical images and information using the DICOM standard. Facilitates communication with devices from different manufacturers.
- DICOM Storage (Send / Receive)
- DICOM Query / Retrieve
- DICOM Basic Print
- DICOM Get Worklist (HIS / RIS)
- DICOM SR viewer
- DICOM Storage Commitment
- DICOM Viewer on CD / DVD
- DICOM MPPS

syngo Expert-i

Enables the physician to interact with the Acquisition Workplace from virtually anywhere in your hospital

Standard and optional system software and applications

IT Security

syngo System Security

Modern way of guarding against malware, viruses and malicious attacks, comprising a bundle of solutions:

- Provides functionality for user management and flexible access control for patient data
- Improves IT security
- Avoids system breakdowns due to malware installations which results in higher system uptimes and reliability
- Reduces risk of unwanted software installations
- Supports local IT personel
- Improves system performance and robustness
- Improves security for the use of external storage devices

myExam Companion

Shared across Siemens Healthineers CT scanners, myExam Companion is an innovative approach to scanner operation and patient-centricity, designed to make work easier for users, personalize procedures for patients, and help deliver consistent and comprehensive results for radiologists. It guides users of varying experience levels through examinations such as stroke, spectral studies, or coronary CTA. myExam Companion asks users the right question at the right time, such as "Is the patient able to hold their breath?", "Does the patient have stents?", or "Are automatic reconstructions of the coronaries needed?" The answers are combined with digital patient data from the FAST 3D Camera, the topogram, ECG, or the radiology information system (RIS) – and then linked to predefined scan parameters. Users of different experience levels can thus individualize and optimize scans for patients and procedures, while keeping radiation and contrast media dose low and patients at ease.

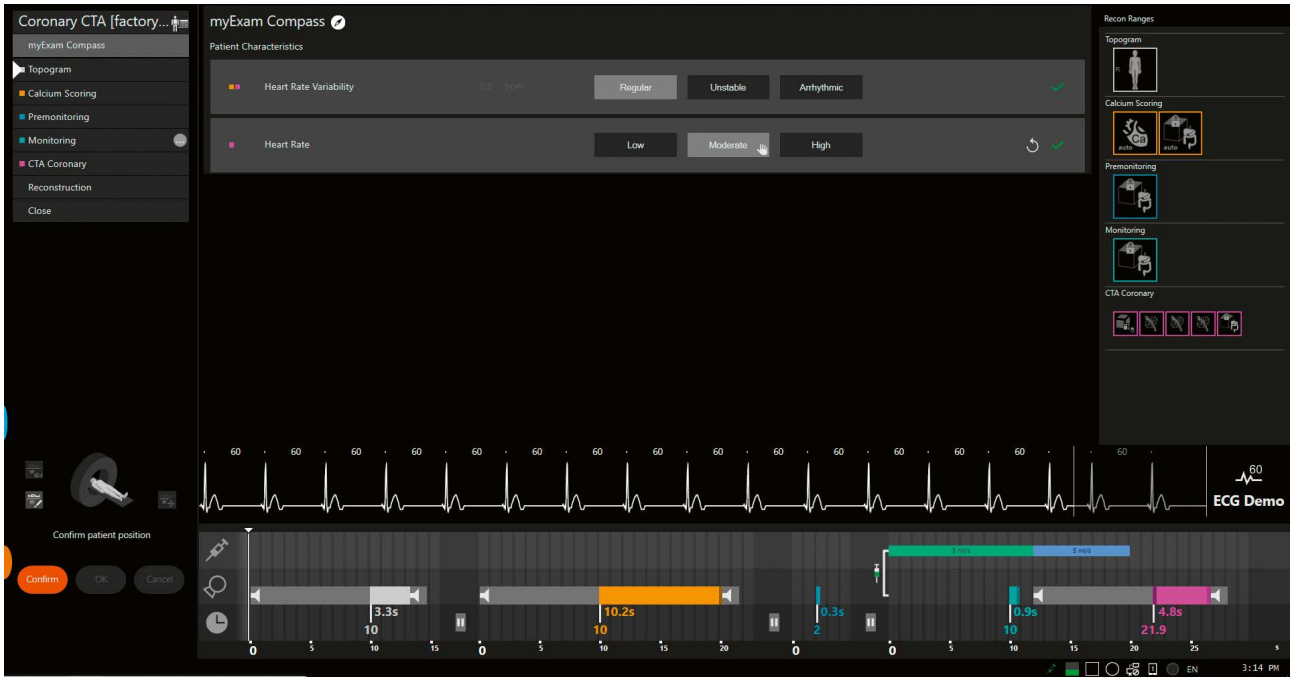
myExam Compass

myExam Compass offers knowledge-based guidance at the hands of the technologist supporting individual patient characterization, based on patient input (size, age, sex, ECG) and interactive questions, adaptable by users, in their own clinical language (e.g., "does the patient have a metal implant?", "can the patient hold the breath longer than 5 sec?").

myExam Compass is based on expert use and condensed knowledge from thousands of exams in our installed base.

Enhance consistency and standardization of your CT procedures by sharing myExam Compass protocols across your institution or other peers through teamplay.

Especially in clinical trials, where protocol consistency is key, myExam Compass helps you harmonize settings for all patients and operators.



myExam Companion

Contrast Medium procedures simplified with easy to answer question user decides to scan WITH or WITHOUT contrast medium by simply selecting an answer, no other manual input needed and the system will automatically take care of adding injector coupling, specific delays, recons and their parameters. A Late Phase scan needed? Simply answer yes to this question and system will apply every setting to acquire a late phase scan.

Cardiac CT made easy independent of operator skills
Especially useful for users less experienced in CT cardiac procedures, myExam Compass suggests which settings are more appropriate for every patient based on the procedure and patient characteristics, finding the optimal combination of acquisition and reconstruction parameters for excellent image quality, standardized results, and always the right dose.

Dual Energy is on when you need it
Especially useful for users less experienced with the DE technique, this holistic approach, powered by myExam Compass, suggests which DE settings are appropriate for which patient based on the procedure and patient characteristics to find the optimal combination of acquisition and reconstruction parameters for standardized results, and always the right examination.

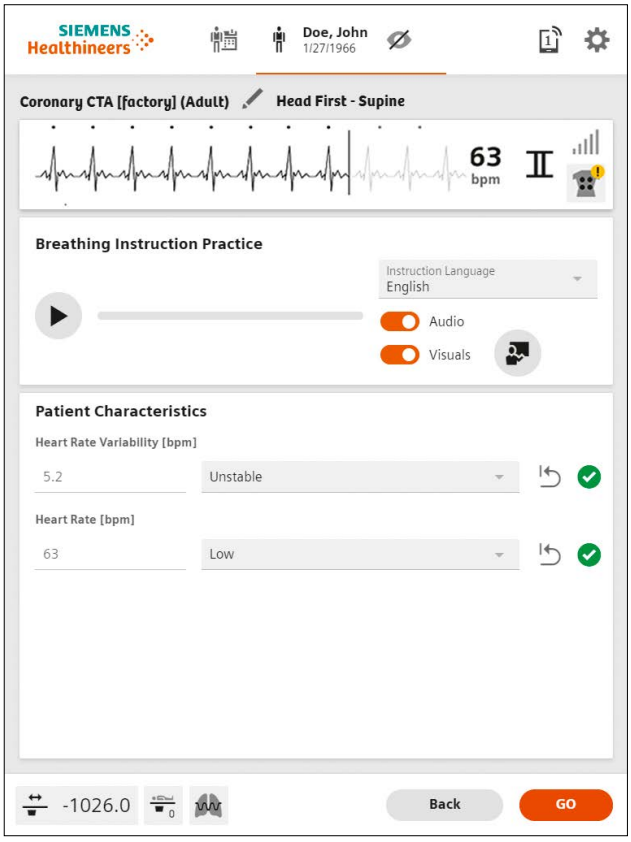
If you are not sure when to use TwinSpiral or TwinBeam or which Recon&GO results to create, just rely on myExam Compass.

Depending on factors such as the patient size, how well they can comply with breathing instructions, or what the indication is, myExam Compass will adapt the protocol settings.

Anticipate potential breathing artifacts
Anticipate potential breathing artifacts by proactively characterizing the patient's ability to comply with an acceptable breathhold. This breathhold characterization is utilized by myExam Compass to adapt scan parameters and optimize scan speed.

myExam Cockpit

The central engine of Exam Compass is driven by this cockpit: the central user interface for fast and intuitive protocol configuration. In this expert mode, users benefit from high flexibility in modifying predefined protocols and the option to integrate their knowledge into standardized protocols, and through myExam Compass, make them available for every user across your institution.



Standard GO technologies

Scan&GO

The operator can reduce walking time and potentially accelerate patient preparation and positioning with the Scan&GO tablet application. At the same time, they can stay close to the patient for most of the examination time.

Post the scan, the operator can preview images after the scan thanks to wireless image transfer to tablet. They can also finalize the exam and trigger pre-configured reconstruction tasks.

With the Scan&GO workflow, the operator can stay mobile and prepare the entire protocol next to the patient in time critical situations. They have the choice to leave the room only when triggering the radiation and spend the rest of the time with their patient.

Check&GO

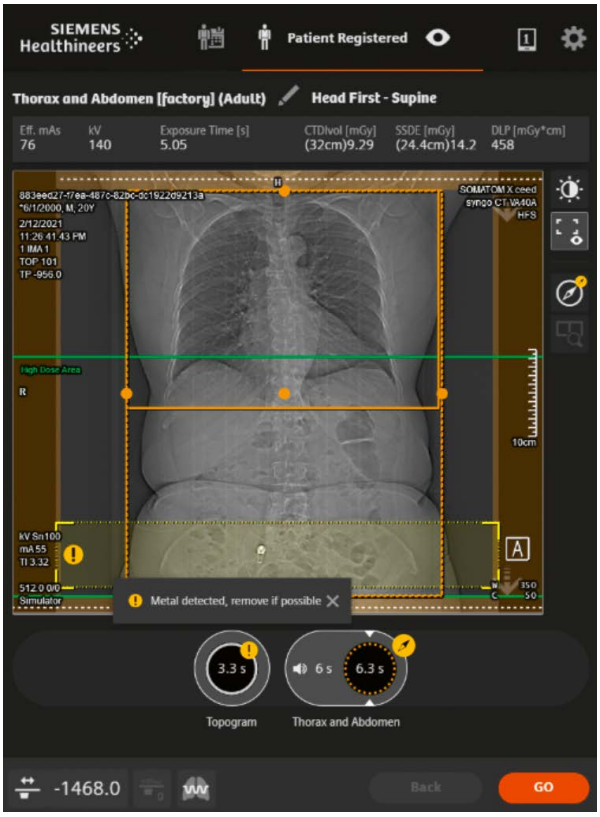
Check&GO is an intelligent algorithm, based on big data, that monitors and flags problems with scan coverage, contrast distribution or the presence of wearable metal objects as they occur, for immediate action or correction. This allows you to correct issues on the go, avoid subsequent errors as well as stop the archival sub-optimal images.

Quality-control images are sent wireless to the tablet, so you can review them directly.

Check&GO Metal Detection helps prevent mistakes and rescans by alerting the user when metallic objects such as keys, belts, chains, earrings are not removed and are present in the scan area after the topogram is done.

Check&GO detects the center and the radius of the arteries, based on different landmarks depending on the scanned body region the arterial enhancement is measured at relevant locations.

Check&GO is available both on the AWP and on the Scan&GO tablet application.



Standard GO technologies

Recon&GO

Recon&GO enables the creation of Inline results, a set of fully automated advanced postprocessing applications as an alternative to the regular *syngo*.via algorithms.

This reduces post-processing to zero-clicks with Recon&GO and its automatically corrected orientations.

Benefit from Recon&GO's standardized and consistent orientations, in typically challenging situations where patients can be mispositioned or uncooperative.

Recon&GO – Inline Results including:

Multi-recon
Automatic generation of multiple series in different orientations (coronal / sagittal / axial) or image impressions (soft tissue / air / bone / ...)

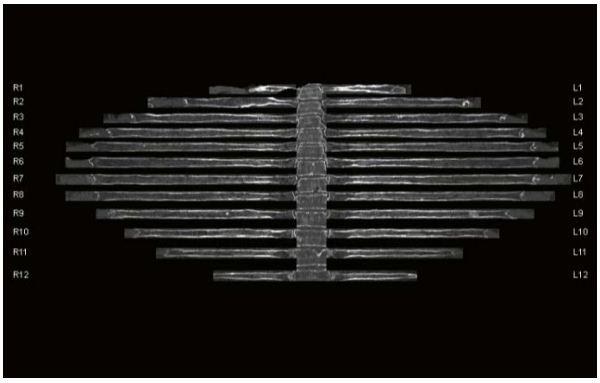
Anatomical Ranges (Parallel / Radial)
Automatic generation of radial and parallel ranges in any anatomical orientation and thickness. This automation saves time by avoiding manual workflow steps. Just configure your required results once and Recon&GO will always create them like a conventional reconstruction.

Table and Bone Removal Radial Ranges
Zero-click bone-free VRT reconstruction that facilitates a precise vascular assessment by visualizing blood vessels without interfering anatomical structures

Vascular Ranges
Zero-click, vessel centerline extraction and anatomical labeling of the main vessels with display of Curved Planar Reconstruction to simplify reporting of findings and stenosis assessment

Spine Ranges
Zero-click reconstruction of anatomically aligned spine reconstructions. The software detects and labels vertebrae within a predetermined scan area, and calculates their position for anatomically correct image reconstructions.

Radial and Parallel Rib Ranges
- Zero-click reconstruction of specific radial and parallel rib visualizations that adapts the rib cage anatomy displaying all ribs spread out in one plane
- Automated rib labeling and numbering



Courtesy of University Hospital Erlangen, Germany

Standard GO technologies

CT View&GO

This viewing application available at the AWP provides you with intuitive and customizable cross-specialty tools for 3D visualization, filming and printing, as well as several post-processing applications.

Customizable user interface, through a Favorite Toolbox

Automatic distribution and filming of images and results

Freely selectable window width and center

Single window

Multiple window settings for multi-image display

Organ-specific window settings, e.g., for soft tissue and bones

Image zoom and pan

Evaluation tools

Parallel evaluation of more than 10 Regions of Interest
- Circle
- Irregular
- Polygonal

Statistical evaluation
- Area / volume
- Standard deviation
- Mean value
- Min. / max. values

Profile cuts
- Horizontal
- Vertical
- Oblique

Distance measurement

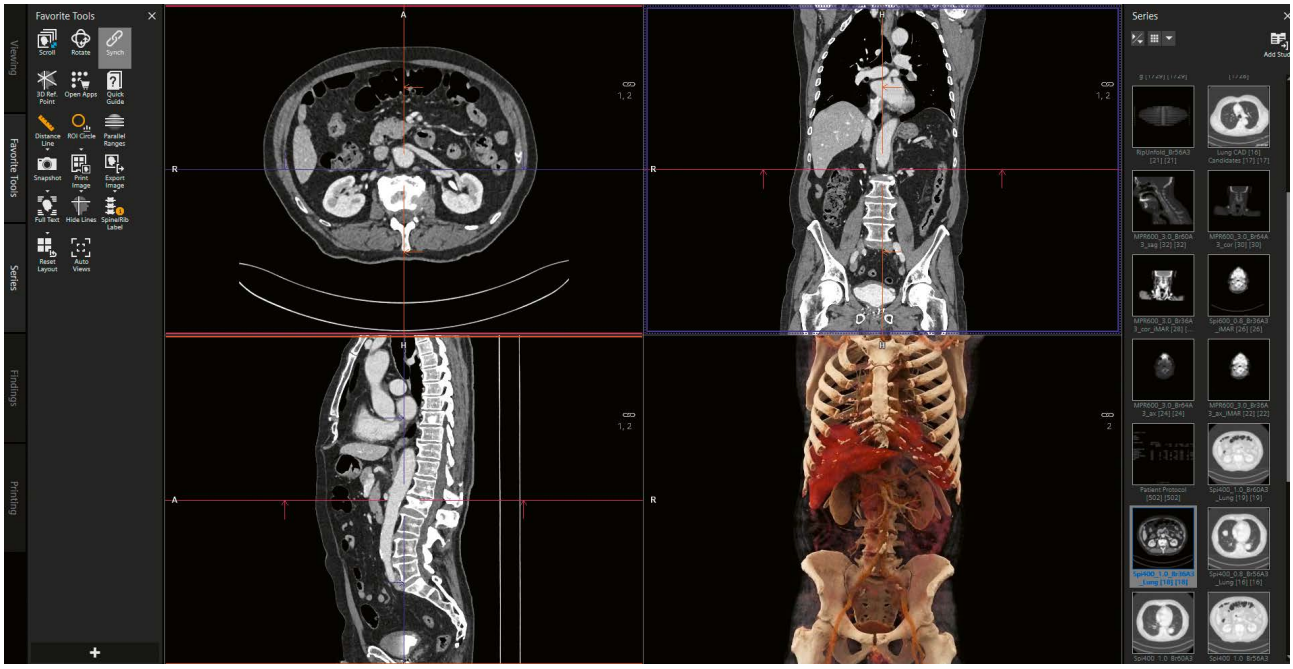
Angle measurement

Online measurement of a 5 x 5 pixel size ROI

Freely selectable positioning of coordinate system

Crosshair

Image annotation and labeling



Standard GO technologies

Filming and printing

Filming
- Digital film documentation, connection to a suitable digital camera
- Connection via DICOM Basic print
- Automatic filming
- Interactive virtual film sheet
- Customizable film formats with up to 64 images
- Filming parallel to other activities
- Independent scanning and documentation
- Freely selectable positioning of images onto film sheet
- Configurable image text

Printing
Documentation on postscript printer supported

3D Visualization

Real-time MPR
- Real-time multiplanar reformatting of secondary views
- Variable slice thickness (MPR thick, MPR thin) and distance with configurable default values
- Viewing perspectives
  - Sagittal
  - Coronal
  - Oblique
  - Double oblique
  - Freehand (curvilinear)

MIP and MinIP
- MIP: Maximum Intensity Projection
- MinIP: Minimum Intensity Projection
- Thin MIP function for projection within a small slab to focus on particular vascular structure

***syngo* VRT (Volume Rendering Technique)**
Advanced 3D application package for the optimal display and differentiation of different organs through independent control of color, opacity, and shading

Integrated functions @ CT View&GO

Table and Bone Removal
Fast accurate presentation of subtracted CT Angiographic data sets

Vessel Extension
- Set of tools and layouts for guided creation of CPR (Curved Planar Reconstructions) for enhanced vascular assessment
- Comprehensive length and diameter measurements

Endoscopic View
Virtual Endoscopy software enabling visualization of airways and intestines

Diameter / WHO area
Longitudinal lesion measurements and WHO for enhanced clinical decisions in oncology

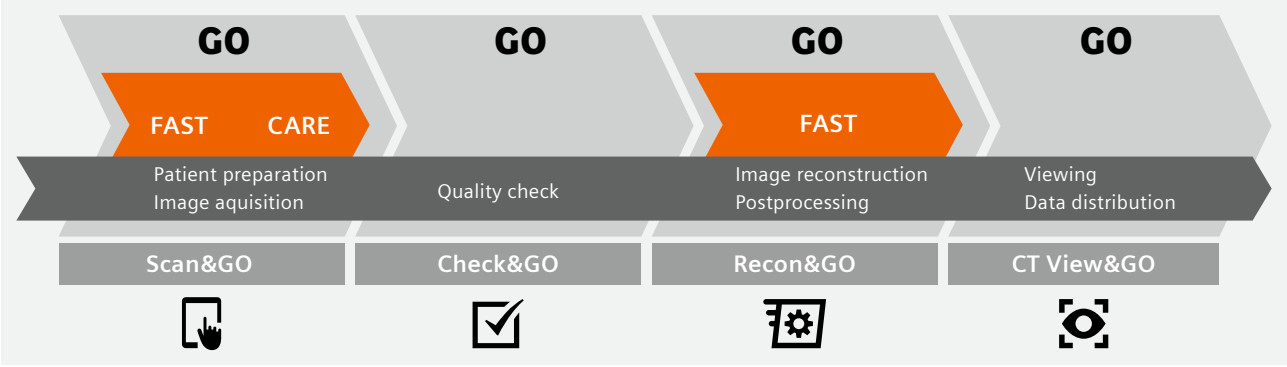
ROI HU Threshold
Evaluation and display tissue densities within a certain HU range

Dual Energy ROI
Basic evaluation of the behavior of different tissues at different energies as an indication of their composition

Lung Nodules Segmentation
The Lung Nodules Segmentation tool in CT View&GO performs an automated segmentation of solid and subsolid lesions in lungs, providing the volume and diameter according to the Lung-RADS guidelines.

Spine Ranges
- Guided reconstruction of anatomically aligned spine
- Curved Planar Reconstructions (CPR)
- Automatic detection and labeling of vertebrae

Average
Allows fusion of two DICOM images on a pixel-by-pixel basis for orthopedic measurements



Standard FAST applications

FAST Planning

AI Machine Learning powered set of algorithms that allow fast, organ-based setting of scan and reconstruction ranges. This enables consistent and reproducible acquisitions in Single and Dual Energy scans. By automating the workflow, users increase efficiency due to reduced manual steps and effort in scan preparation.

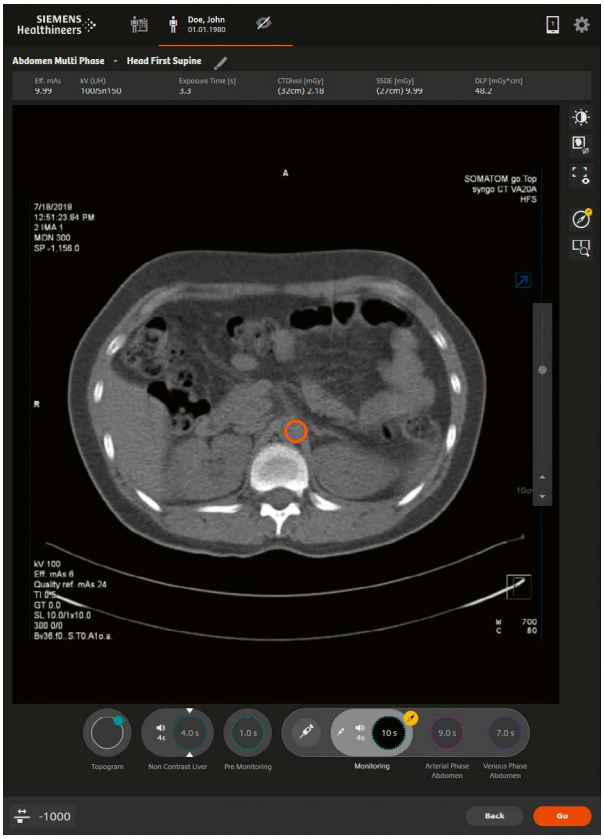
This Machine learning algorithm is trained with several hundreds of patient datasets in order to overcome even the most challenging anatomies (e.g., bypass). Landmark detection technology recognizes known "human anatomy anchors" on the topogram and the scan range automatically snaps to the correct region.

It prevents the range from being set too short or too long, so no parts of the organ are cut off or over-radiated.

In addition to the Acquisition Workplace (AWP), FAST Planning is also integrated in the Scan&GO tablet user interface.

FAST ROI

Automatic ROI identification for the aorta and the pulmonary trunk for optimal enhancement timing



FAST Adjust

FAST Adjust assists the user to handle system settings in a fast and easy way by automatically solving of conflicts by one single click

FAST Contact[1]

FAST Contact is the easiest way to contact our service experts directly from the scanner console for technical and clinical application support. teamplay Fleet – our fleet management tool – also tracks and archives service tickets generated with FAST Contact.

[1] *teamplay Fleet and FAST Contact are subject to country-specific availability*

Optional FAST applications

FAST 3D Camera

The AI-powered FAST 3D Camera enables an automated workflow to safeguard precision and consistency in patient positioning – enabling high efficiency, increased image quality, and an optimized isocenter for an optimal dose, regardless of individual skills.

The algorithms of the FAST 3D Camera support accurate and reproducible positioning based on 3D image and infrared measurements, which even recognize body contours, for example, when people are wearing thicker clothes. The following specialized applications are included:

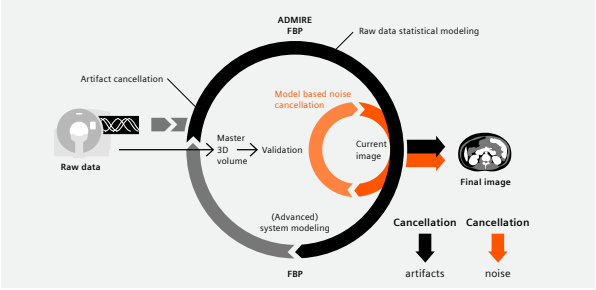
- FAST Isocentering, at the push of a button, provides the correct isocenter position, enabling the right dose modulation and consistent images.
- FAST Range supports scanning the correct body region with no cut-off – by aligning the automatically identified anatomical position with the protocol.
- FAST Direction helps safeguard the right scan direction, which is crucial when moving the table with infused patients.
- FAST Topo enables faster scan speeds in topograms, which prevents breathhold artifacts. It also has the potential to decrease the topogram dose.

The smart communication between the tablet and the FAST 3D Camera helps reduce mistakes even with noncooperative patients between planning and scanning the topogram, thanks to reactive algorithms that will adapt the topogram planning even if patients move.

Available as ceiling-mounted option.

Standard CARE applications

Advanced Modeled Iterative Reconstruction (ADMIRE)[1]



With ADMIRE – Siemens Healthineers' Advanced Modeled Iterative Reconstruction – clinical images additionally benefit from higher resolution at organ borders and improved edge delineation.

As demonstrated using SOMATOM Force CT data, ADMIRE may simultaneously enable[1]

- 80% to 85% dose reduction at the same image quality, and
- 73% to 77% image noise reduction at the reduced dose, and
- up to 42% improved high-contrast spatial resolution at reduced dose and reduced image noise compared to images reconstructed with WFBP[2].

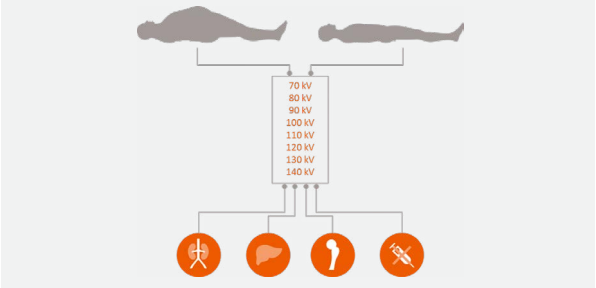
Alternatively, ADMIRE may enable[1]

- up to 150% improved low-contrast detectability (factor 2.5) at the same dose, or
- up to 90% image noise reduction at a constant dose, or
- up to 87% improved high-contrast spatial resolution at 85% reduced dose and constant image noise, or
- up to 38% improved high-contrast resolution at 90% reduced image noise and constant dose.

[1] *Image quality as defined by low-contrast detectability using a model observer method for evaluation. As demonstrated using SOMATOM Force data, equivalent low-contrast detectability can be achieved with 80% to 85% less dose using ADMIRE at the highest strength level for thin (0.6 mm) reconstruction slices in measured and simulated body and head phantoms for low-contrast objects with different contrasts. In clinical practice, the use of ADMIRE may reduce CT patient dose depending on the clinical task, patient size, anatomical location, and clinical practice. A consultation with a radiologist and a physicist should be made to determine the appropriate dose to obtain diagnostic image quality for the particular clinical task.*

[2] *Weighted Filtered Back Projection*

CARE kV



CARE kV automatically tailors tube voltage according to patient size and clinical task.

Simplify processes by automatically aligning mAs with the kV setting.

With the selection of optimal kV level between 70 and 150 kV, CARE kV minimizes dose. It further simplifies the process by automatically aligning the tube current with the selected kV.

10 kV Steps

Adapt your kV selection more precisely to reduce radiation dose at maintained image quality for a broad range of patient sizes.

Benefit from patient-specific and user-independent selection of the optimal kV setting with CARE kV in 10 kV steps.

More patient-specific dose management thanks to finer kV selection in 10 kV steps for individual dose management

CARE Child

CARE Child offers scan parameters to be adapted to even small patient sizes. Dedicated pediatric protocols automatically set a low tube voltage – in most cases 70 kV – while CARE Dose4D optimizes dose distribution and offers special modulation curves.

CARE Filter

Specially designed X-ray exposure filters installed at the tube and the collimator for protocol-specific optimization of patient dose and image quality

Permanent filtration of X-ray tube assembly
Equivalent to 6.8 mm Al @ 140 kV

Tube collimator

- Equivalent to 0.5 mm Al in the isocenter
- 1 mm Al with cardio wedge

Standard CARE applications

CARE Bolus CT

Scan mode for contrast bolus-triggered data acquisition

The procedure is based on repetitive low-dose monitoring scans at one slice level and analysis of the time density curve in an ROI (Region of Interest).

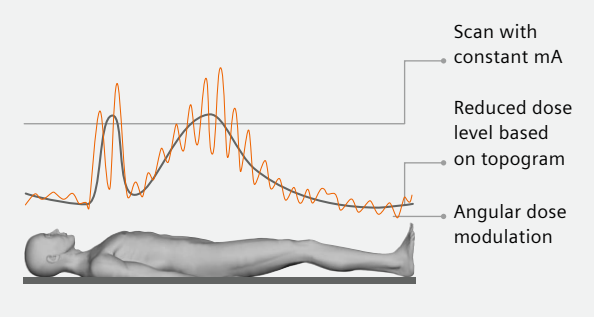
CARE Bolus CT allows the planning and the execution of contrast workflows within the Scan&GO user interface.

CARE Topo

Real-time topogram

Manual interruption possible once desired anatomy has been imaged

CARE Dose4D



Fully automated dose modulation solution. The algorithm automatically modulates tube current for optimum image quality.

This results in deduced dose levels, depending on patient size and anatomy, i.e. there is automatic patient- and organ-specific tube current adaption

X-CARE

Provides organ dose reduction for radiation-sensitive peripheral organs, e.g., eye lenses, while maintaining image quality

Keeps the average $CTDI_{vol}$ constant, i.e. with and without X-CARE

myExam Companion individualizes the utilization of X-CARE by considering the gender and breathhold capability of the patient

Flex Dose Profile

For long scan ranges, Flex Dose Profile works in combination with CARE Dose 4D and FAST Planning to allow a more optimal modulation of the dose.

In longer scans, some organs require more dose than the rest of the scan, i.e. there are different target dose levels needed for different anatomical regions, e.g., in regular thoracoabdominal examinations or in chest pain or TAVI procedures. FAST Planning automatically detects individual patient landmarks and anatomies, while Flex Dose Profile adjusts the tube currents for more personalized and accurate dose handling.

Flex Dose Profile is displayed on the AWP and the Scan&GO tablet with the same visual logic as any other procedure, so users of any level of experience can utilize it right away.

Protocol password protection

Prevent unauthorized access to scan protocols and avoid unauthorized modifications.

DICOM SR Dose Reports

DICOM structured file allows for the extraction of dose values ($CTDI_{vol}$, DLP) to create transparency and document dose values.

DoseLogs

Whenever the set reference dose levels are exceeded automatically, a report is created on the system. The report can be used for audit purposes, for example.

Dose Notification

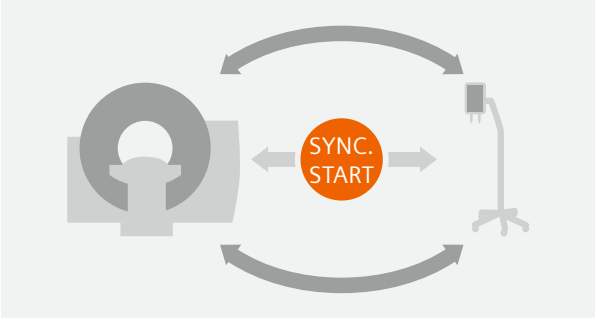
The software checks the dose values per chronicle entry. May help to protect from over-radiation and warn the operator if set dose thresholds are exceeded.

Dose Alert

The software checks the accumulated dose per z-position. May help to protect from over-radiation and warn the operator if set dose thresholds are exceeded.

Optional CARE applications

CARE Contrast III



CARE Contrast III increases process efficiency and standardizes quality of care with technology that lets you synchronize and manage CT scan and contrast media injection, with the interchange of contrast injection protocols (including parameters like phase, flow, duration, volume).

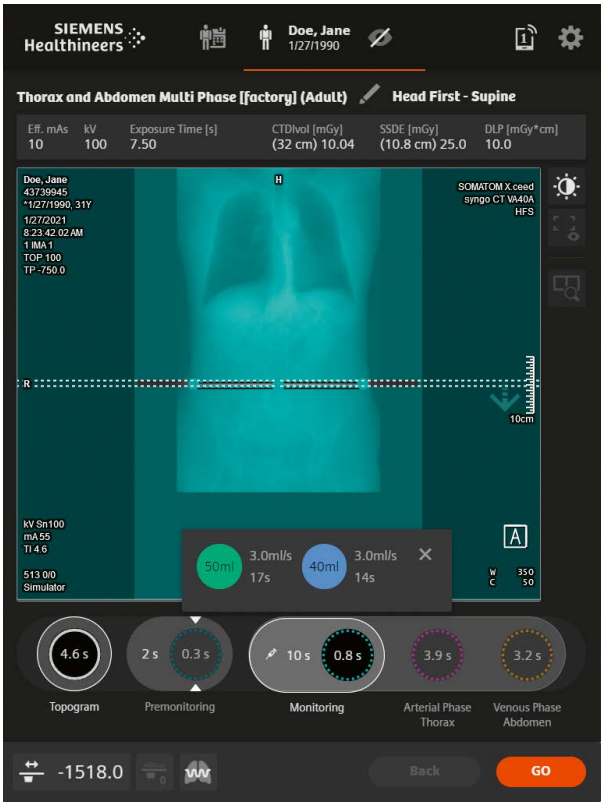
Define and manage contrast media protocols on the scanner console.

Combined scan and contrast media protocols

Transfer contrast media protocols from the scanner console to the injector.

Transfer contrast media protocol to patient protocol.

Selected pre-defined factory protocols[1] including quantified parameterization of flow and concentration for the contrast media, calculated for the average patient



[1] *The FDA does not endorse the default factory contrast protocols over the range of drug administration options provided in the drug label*

Optional system software and packages

iMAR

iMAR (iterative Metal Artifact Reduction) reduces metal artifacts for better image quality with no increase in dose.

The high-end algorithm can handle a wide variety of metal implants. By reducing metal artifacts, it improves visualization of soft tissue. It even allows you to address more challenging cases, such as those involving dental fillings, coils, implants, and pacemakers. Since metal can often be an issue in trauma cases, our iMAR algorithm is a key advantage for this clinical field too.

iMAR reconstructions should be performed and evaluated in combination with standard reconstructions.

iMAR can be combined with TwinSpiral and TwinBeam Dual Energy acquisition, as well as with ADMIRE iterative reconstruction to further reduce dose.

iMAR supporting bone kernels ≤ 56

Cardiac Imaging

The Cardiac Imaging Package allows for comprehensive cardiac assessment and clinical consistency in cardiac CT with ease. Optimized, fully tablet-operated scan preparation, fast scanning, and standardized results in every cardiac case enabled by the integrated GO technologies allow you to devote more time to your patient.

Especially useful for users less experienced in cardiac CT procedures, the exclusive myExam Companion suggests which settings are more appropriate for every patient based on the procedure and patient characteristics and finds the optimal combination of acquisition and reconstruction parameters. By measuring heart rate and rhythm, the system automatically chooses the most appropriate phase of the heart cycle to scan and later reconstruct.

Physiological Measurement Module
Three-channel ECG cable connection. The ECG signal is automatically checked for impedance and monitored on the tablet.

ECG-triggered scan modes
Adaptive prospective (Cardio Sequence) and retrospective (Cardio Spiral) ECG-triggered scanning to obtain CT images of the heart in defined phases of the cardiac cycle at a minimum rotation time of 0.3 s

Native temporal resolution of down to 150 ms

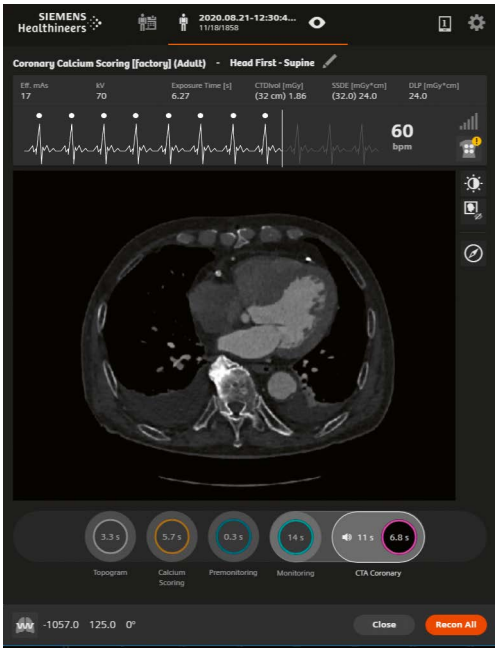
Temporal resolution down to 75 ms (with Bi-Segment)

Adaptive ECG-synchronized dose modulation (pulsing) allowing additional dose savings

Cardio BestPhase
Cardiac BestPhase is a dedicated software which automatically calculates and detects the optimal phase for motion-less coronary visualization. The phase is defined as either end-systole, end-diastole or both timepoints and is automatically reconstructed.

ZeeFree
is a detector-width-independent cardiac reconstruction feature which allows the reconstruction of ECG-gated spiral or ECG-triggered sequence data with alignment of stacks originating from separate cardiac cycles or patient breathing. This is realized by reconstruction of separate stack of images for each contributing cardiac cycle, followed by a non-rigid registration step based on the available overlapping image data between neighboring cardiac cycles. The feature is a scanner integrated reconstruction option, which can be manually enabled or disabled by the user. ZeeFree works independent from the physical detector width of the acquired data.

Any kV CaScoring
Calcium scoring can be performed either at any kV or with Tin Filter enabling Agatston equivalent low-dose coronary calcium scoring.



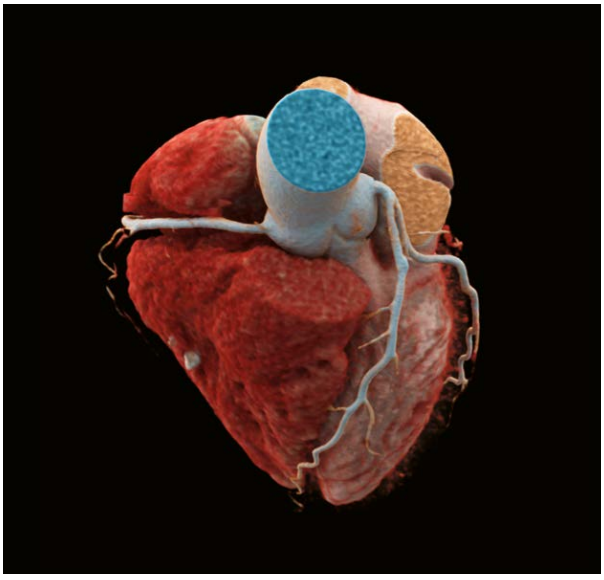
Optional system software and packages

Recon&GO

- Inline Results – Cardiac Ranges
- Inline Results – Vessel Ranges (LAD, RCA, CX)
- Inline Results – CaScoring
- Inline Results – Heart Isolation
- Inline Results – Coronary Tree

After the scan, Recon&GO produces ready-to-read results for instant evaluation. Zero-click CPR of the main coronaries and radial VRT ranges of the coronary tree (as recommended in the SCCT guidelines 10) help you quickly rule out coronary artery disease.

Inline CaScoring makes the Calcium Score available as zero-click reconstruction. With the known functionality of Recon&GO, Inline CaScoring calculates automatically the total Agatston Score as well as the Coronary Age (based on trial data) and archives them directly in the PACS. Results can be opened in *syngo*.CT CaScoring directly at the AWP and further processed if needed.



Courtesy of Clinica Universidad de Navarra, Pamplona, Spain

CT View&GO

CT View&GO provides a set of tools for basic coronary CTA evaluation on the AWP including automatic heart and coronary tree isolation and dedicated layouts for guided creation of CPR (Curved Planar Reconstructions) for enhanced vascular assessment and stenosis measurement.

syngo.CT CaScoring application for AWP

syngo.CT CaScoring allows visualization and quantification of calcified coronary lesions volume (in mm^3), calcium mass (mg calcium hydroxyapatite), vessel-specific and total Agatston equivalent score and the number of lesions. Scoring can be performed separately for the main coronary branches (RCA, LM, LAD, CX). In addition, it calculates the virtual coronary age by comparison against a reference group. Combined with Rapid Results Technology it enables zero-click post-processing of both Agatston Scoring as well as coronary age analysis.

Optional system software and packages

Spectral Imaging with Dual Energy

By allowing you to characterize, highlight, and quantify different materials this produces rich diagnostic information that a conventional single source scan cannot deliver. It does this without dose penalty in comparison to a standard 120 kV scan, and even allows you to further minimize radiation with any of our existing dose-reduction technologies.

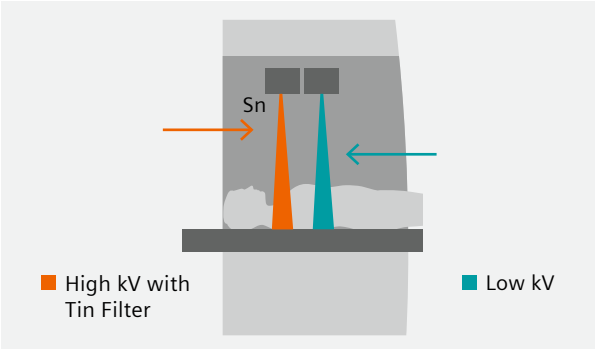
Dual Energy procedures are handled on the AWP or the Scan&GO tablet with the same visual logic and automation as any other procedure, so users of any level of experience can perform them right away.

Especially useful for users less experienced with the DE technique, this holistic approach, powered by myExam Compass, suggests which DE settings are more appropriate for every patient based on the procedure and patient characteristics and finds the optimal combination of acquisition and reconstruction parameters for excellent image quality, standardized results, and always the right dose.

TwinSpiral Dual Energy[1]

A new holistic solution for spectral imaging is introduced. The TwinSpiral scan mode offers the possibility to acquire two consecutive spiral data sets at different energies used for non-contrast scans and the two different kV levels with independent mAs modulation deliver a combination of both morphological and functional information within one examination.

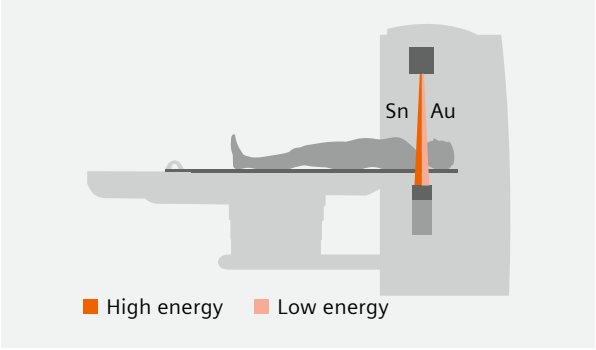
The robustness of the spectral separation is a key factor for the quality of the final images. The spectral properties of the Tin Filter lead to better spectral separation and therefore, amongst other benefits, potentially results in better tissue characterization. Leverage the power of the Mobile Workflow with end-to-end Spectral imaging protocols running from the tablet. Now users of any level of experience can perform spectral imaging examinations with dual energy from the tablet right away, as the TwinSpiral protocols offer a holistic approach which starts from the scanning part and extends to the automatic way of generating the results. The new TwinSpiral workflow feels like a single scan. The patient experiences virtually zero inter-scan delay between the scans. Empowered by such a holistic approach, overall protocol execution time is expected to be reduced.



[1] *TwinSpiral Dual Energy supports the following Dual Energy applications: syngo.CT Dual Energy (incl. Monoenergetic), syngo.CT DE Gout, syngo.CT DE Calculi Characterization, syngo.CT DE Bone Marrow and syngo.CT Monoenergetic Plus (not for visualization of iodine, only for metal artifact reduction)*

Optional system software and packages

TwinBeam Dual Energy[1]



TwinBeam Dual Energy is an innovative, high-end approach that allows simultaneous acquisition of high and low kV spectra in a single spiral scan mode.

To create two X-ray spectra (high and low) simultaneously from one tube, the powerful Vectron tube assembly generates a prefiltered X-ray beam before it reaches the patient. Spectral separation is achieved with the integrated gold and tin filter.

No dose penalty in comparison to a standard 120 kV scan
Minimize dose by automatic selection of optimal kV between AuSn120 and AuSn140 with CARE kV and all available dose reduction technologies such as CARE Dose4D and ADMIRE.

Image acquisition is possible for all rotation times. High and low energy image series are reconstructed separately. Composed reconstruction delivers a single energy image dataset using the full information by directly combining the low and high energy data from the detector. With the full number of projections available for both spectra, there is no compromise on image quality.

Composed reconstruction enables single-energy image dataset with diagnostic image quality

[1] *TwinBeam Dual Energy supports the following applications: syngo.CT Dual Energy – Optimum Contrast, syngo.CT DE Gout, syngo.CT DE Calculi Characterization, syngo.CT DE Direct Angio, syngo.CT DE Lung Analysis, syngo.CT DE Virtual Unenhanced (contains Liver VNC), syngo.CT DE Monoenergetic Plus (mainly for visualization of iodine, basic metal artifact reduction), syngo.CT DE Hardplaque Display*

Recon&GO

Experience a zero-click Dual Energy workflow automated by DE inline results

Inline Results – DE SPP (Spectral Post-processing)
Improve workflow by fusing low- and high-energy data into the SPP data format.

This reduces the amount of data and transfer times to PACS. Improve workflow by fusing low- and high-energy data into the SPP data format. Enables on-the-fly processing of Dual Energy data.

Use the DICOM-conformant spectral imaging data format *SPP* to improve the workflow of applications utilizing DE information.

Reduce file size by bundling L+H datasets into a conventional image (e.g., C-type or mixed) or into a monoenergetic image (e.g., 40/50/70 keV).

Recon&GO – Spectral Recon
Benefit from the speed of Spectral Imaging reconstructions calculated directly at the IRS based on the spectral information available in the raw data, while reducing the amount of data and transfer times to PACS.

Handle Spectral Imaging results seamlessly, evaluate and manipulate these volumetric data in 3D, create additional reconstructions or measurements, as in any other standard series:
Full freedom for further multiplanar reconstructions and quantification of DE related parameters.

Spectral Recon includes:
- Mixed
- Monoenergetic Plus
- VNC (Virtual Non-Contrast)
- Iodine map

Inline Results – DE Ranges
Recon&GO standardizes results by automatically generating the most appropriate DE inline results in any required orientation and thickness. This automation saves time by avoiding manual workflow steps. Just configure your required results once and Recon&GO will always create them like a conventional reconstruction.

Optional system software and packages

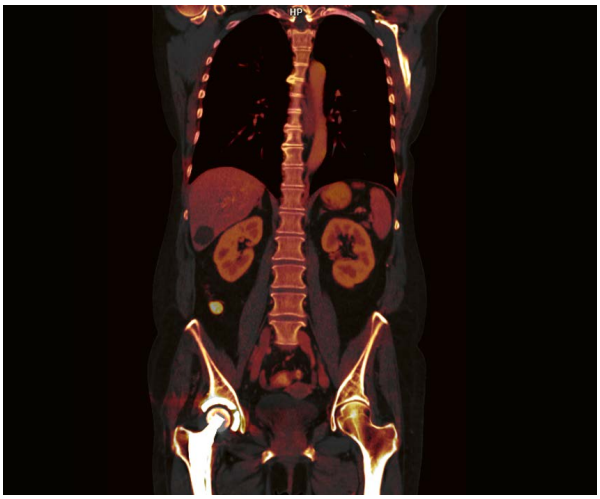
Inline Results includes the following DE Ranges:
- Mixed
- Monoenergetic Plus
- Direct Angio (bone removal)
- VNC (Virtual Non-Contrast) including Liver VNC and Brain Hemorrhage
- Iodine map
- Lung PBV (Perfused Blood Volume) / Vessels
- Gout
- Calculi Characterization
- VNCa (Virtual Non-Calcium) including Bone Marrow
- Rho/Z (Electron density / atomic number)[1]
- Fat Map
- Brain Hemorrhage
- Hard Plaque
- Bone Marrow

CT View&GO

As an all-in-one, cross-specialty viewing solution, CT View&GO provides tools for routine DE evaluation in different clinical fields

CT View&GO – Dual Energy ROI
Basic evaluation of the behavior of different tissues at different energies as an indication of their atomic composition

CT View&GO – Interactive Spectral Imaging
- Interactive Spectral Imaging makes Dual Energy as easy as windowing – use additional spectral information at a glance.
- It enables loading and use of the most common Dual Energy application classes (Mixed, Monoenergetic Plus) directly within CT View&GO by the click of one button.



Courtesy of University Hospital Erlangen, Germany

syngo.CT Dual Energy applications for AWP

Advanced applications for Dual Energy evaluation and postprocessing on the AWP

***syngo*.CT Dual Energy**
- Preparing and viewing of Dual Energy data
- Mixed image calculation
- Monoenergetic
- Optimum Contrast
- Rho/Z (electron density / effective atomic number)[1]

***syngo*.CT DE Gout**
Color-coded visualization of deposited uric acid crystals in peripheral extremities

***syngo*.CT DE Calculi Characterization**
- Visualization and characterization of kidney stones
- Seamless navigation through the visualized stones

***syngo*.CT DE Brain Hemorrhage**
Distinguish hemorrhage from contrast agent

***syngo*.CT DE Direct Angio**
- Highlighting of bone structures
- Subtract bones with a single click

***syngo*.CT DE Hard Plaque Display**
Preprocessing and visualization of hard plaques. Differentiation of calcified plaque and iodine contrast media with color-coding

***syngo*.CT DE Lung Analysis**
- Combination of *syngo* Lung PBV and *syngo* Lung Vessels
- Lung isolation
- Visualization and quantification of the local iodine concentration in the lung parenchyma and vessels

***syngo*.CT DE Bone Marrow**
Color-coding of bone marrow

***syngo*.CT DE Virtual Unenhanced**
- Lodine uptake quantification
- Calculation of virtual unenhanced image
- Calculation of fat map in the liver

***syngo*.CT DE Monoenergetic Plus**
- Compare lesions and tissues by displaying multiple ROIs and associated attenuation curves
- Display monoenergetic images in the range of 40–190 keV

[1] *Not cleared for use as a basis for radiation therapy planning, but for visualization only*

Optional system software and packages

LungCAD

Recon&GO

Inline Results – LungCAD
Zero-click LungCAD (Computer Aided Detection) series

Reconstruction, designed as PACS-Ready second reader tool to assist radiologists in the detection of pulmonary nodules during review of CT examinations of the chest.

Designed to assist in the detection of solid pulmonary nodules

CT View&GO

LungCAD
As an all-in-one, cross-specialty viewing solution, CT View&GO provides a LungCAD tool, as computer assisted second reader solution for evaluation on the AWP.

LungCAD potentially makes results more objective and consistent, and shortens the radiologists' learning curve to accommodate varying skill levels of physicians in interpreting diagnostic images.

Emergency Imaging

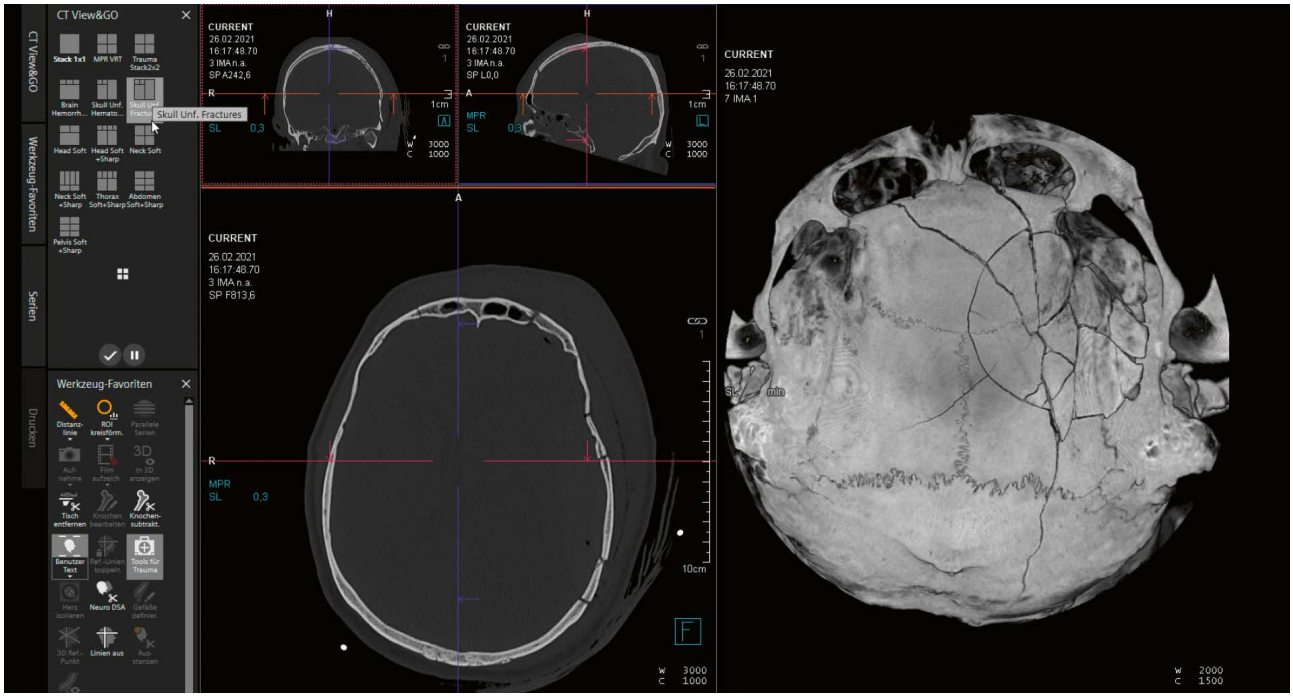
With dedicated CT View&GO and Recon&GO applications workflow in emergency procedures is quicker and easier. If myExam Satellite is available, the same applications are accessible there for direct evaluation of images without interrupting the scanning program.

Recon&GO

Inline Brain Hemorrhage
Automatic detection and PACS notification of suspected intracranial hemorrhage

Inline Brain Hemorrhage SAH
The application offers automatic and AI-based triage of potential subarachnoid hemorrhage cases with the Recon&GO Inline Results. This helps in detecting potential intracranial hemorrhage/edema cases including Subarachnoid space.

Inline Skull Unfolding
Automatic curved MIP images of skull and brain surface to support detection of skull fractures and thin surface hematoma



Optional system software and packages

Inline Brain Hyperdensity
The Brain Hyperdensity functionality assists in segmenting and quantifying hyperdensities in the brain on head noncontrast CT scans. The functionality is part of *syngo* .CT Brain Quantification and it is available through Recon&GO Inline Results.

Results can be automatically sent to PACS

CT View&GO

Trauma layouts
Predefined layouts are automatically loaded and filled with corresponding data for head, neck, thorax and abdomen and pelvis. Data from Skull Unfolding and Brain Hemorrhage is automatically displayed within the layouts.

Neuro Imaging

The Neuro Imaging Package provides you with a tiltable head holder and various tools for assessment of stroke and other neurological diseases: native neuro scanning with automatic ASPECTS, neuro DSA (Digital Subtraction Angiography) and neuro perfusion.

Tiltable Head Holder
- Tiltable carbon fibre Head Holder for the fixation of the patient's head
- Tilt range between +30 till –15 degree

Flex 4D Spiral – Neuro[1,2]

Flex 4D Spiral Neuro adds functional information to morphology, with perfusion range larger than the detector width up to 10 cm, thus easily covering the entire supratentorial brain.

Facilitates dynamic angiographies for head and neck with a coverage of up to 30.5 cm [3]

These dynamic procedures are handled on the AWP with the same visual logic as any other procedures, so users of any level of experience can perform them right away.

CARE Technologies are also available for dynamic studies in order to help follow the ALARA principle, for instance with flexible adaptation of the scan range to the region / organ of interest, low kV modes (70, 80, 90 kV), CARE kV and CARE Dose 4D.

CT View&GO

Stroke Layout
The Stroke Layout will automatically load stroke results in a dedicated layout to facilitate the readability of the results. This layout will show the relevant patient results, based on the type of stroke (e.g., in case of Ischemic Stroke). The results can be read from CT View&GO at the scanner and/or MM Reading (*syngo*.via).

Neuro DSA
The Neuro DSA tool within CT View&GO provides a bone-free view of the cerebral vasculature based on the subtraction of an additional non enhanced CT (NECT) scan that is three-dimensionally registered to the CTA data set. The improved visualization of vascular structures in the skull base area will help to delineate aneurysms and other vascular diseases. A negated Image provides an angio-like view.

Recon&GO

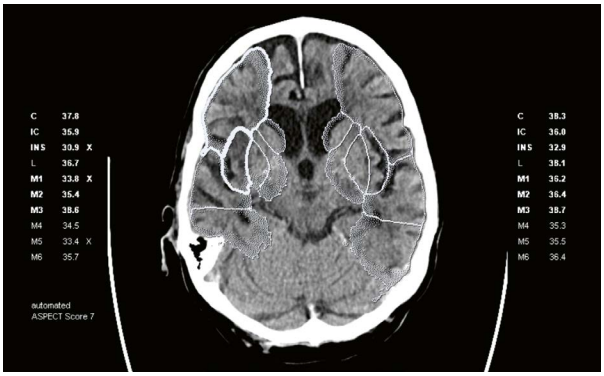
Inline Results – ASPECTS
Zero-click ASPECTS (Alberta Stroke Program Early CT Score) for assessment of early ischemic changes over non-contrast head CT scans.

The affected brain regions are automatically highlighted as an overlay on the CT image based on a quantitative topographic comparison of the left and right hemisphere. Automatic segmentation into 10 cerebral regions based on imaging data of patients who presented with symptoms of acute stroke with proven vascular occlusion.

The images and results are automatically calculated in the background and can be directly sent to PACS without any user interaction. This makes Inline ASPECTS routine-ready by providing consistent results independent of the user and always available especially in urgent situations when time is a scarce resource.

[1] *Not compatible with ICS X. Basic, IRS X. Basic*
[2] *Not compatible with 227 kg table*
[3] *Optional with MPT Table*

Optional system software and packages



Inline Results – Neuro Perfusion
Neuro Perfusion supports the assessment of brain tissue perfusion through a contrast CT head scans with a full automatically and reproducible quantitative grading system for tissue differentiation, i.e., whether Penumbra or core infarct. Recon&GO provides inline neuro perfusion calculation and automatic transfer to PACS.

syngo.CT Neuro Perfusion application for AWP

syngo.CT Neuro Perfusion offers a fully automated or guided workflow to evaluate areas of brain perfusion. The guided workflow is a five step process to view the perfusion parameters including core infarct and penumbra.

syngo.CT Neuro Perfusion provides the following quantitative 3D images based on the deconvolution model and the maximum slope model:

- Cerebral Blood Flow (CBF) image
- Cerebral Blood Volume (CBV) image
- Mean Transit Time (MTT)
- Time to Peak image (TTP, image of bolus peak time)
- Time to Start image (TTS, image of bolus arrival time)
- Time to Drain (TTD)
- TMax (Transit time to the center of the Impulse Response Function)
- Flow Extraction Product (Permeability)

4D Imaging[1,2,3]

Flex 4D Spiral – Body

Continuously repeated bidirectional table movement during spiral acquisition enables an extended range for 4D information.

Facilitates volume perfusion studies in body applications for a perfusion range of up to 21.5 cm[4]

Facilitates dynamic studies up to a scan range of 48 cm[4]

These dynamic procedures are handled on the AWP with the same visual logic as any other procedure, so users of any level of experience can perform them right away.

Precision Matrix

Reconstructions of images with matrix sizes of up to 1024 x 1024 and 768 x 768, useful to keep spatial resolution high even at full scan FOV

Powered by myExam Companion, the right image matrix size for axial and 3D reconstructions is automatically selected depending on FOV offering a balance between storage demand, reconstruction time and spatial resolution.

[1] *Not compatible with ICS X. Basic, IRS X. Basic*
[2] *Not compatible with 227 kg table*
[3] *Recommended applications for evaluation in syngo.via: syngo.CT Body Perfusion and syngo.CT Dynamic Angio*
[4] *Maximum perfusion and dynamic ranges available with multi-purpose patient table. 19.5 cm body perfusion range and 44 cm dynamic range available for 307 kg patient table.*

Optional system software and packages

CT Dental

syngo.CT Dental application for AWP
syngo.CT Dental supports pre-surgical planning for dental operations by reformatting of curved panoramic and paraxial views along the jaw-bone, as well as, definition of the mandibular canal.

Filming in true anatomical size allows direct measurement based on the x-ray films.

CT Osteo

CT View&GO – CT Osteo
- View and save the results of Bone Mineral Density [BMD] in mg CA-HA/ml of the right and the left sides of the cortical bone and the trabecular bone and total BMD (in a tabular form).
- See the T-Score and the Z-score in a graphical form based on a default database of a Caucasian population or a user defined database.

ESP Phantom
European Spine Phantom for standardization of Osteo CT measurements

Optional system software and hardware for CT-guided interventions

myNeedle Companion for CT-guided interventions

On the SOMATOM X.cite we are integrating the first solution that harmonizes planning and guidance for percutaneous needle procedures across modalities. myNeedle Companion supports the interventionalist by utilizing the standard system tablet to interact with the system software and the images with touch-gestures from inside the examination room to stay close by the patient during the procedure. In addition the intervention user interface myNeedle Guide is displayed on a 24" or 32" in-room monitor either mounted at the ceiling or on a cart.

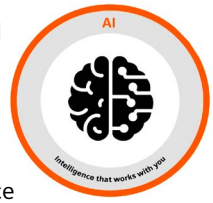
myNeedle Guide 2D[1]

Assists you in planning and guiding the needle during in-plane percutaneous CT-guided interventions. Dedicated tools support the planning of a needle path by providing distance and angle measurement from the target to the needle entry point in one or several axial CT slices. It includes i-Sequence scan mode referred to as FAST i-Sequence as it allows for quick scan repetitions, e.g., for dynamic monitoring of the needle.

myNeedle Guide 3D[1]

Supports all kind of percutaneous procedures, from simple in-plane interventions, to complex, double-angulated procedures. myNeedle Guide 3D supports planning of multiple needle paths by measuring distances and angles from the target to the needle entry point on one or several axial CT slices and as well on Multi Planar Reconstructions. To leverage the optimized image for planning the needle path 3D images from other modalities or prior CT scans can be fused with the actual CT scan.

It includes:



- **myNeedle Detection:** an AI based algorithm to detect the needle. The software scrolls automatically to the axial image with the needle tip and aligns the MPR views to the needle. In addition the distance to the planned target point and the angular deviation is calculated.
- i-Sequence scan mode referred to as FAST i-Sequence as it allows for quick scan repetitions, e.g., for dynamic monitoring of the needle, either with a volume coverage of 3 × 3 mm or 3 × 5 mm for inplane procedures or covering the full detetcor width which allows for 3D planning and guidance.
- i-Spiral mode for flexible volume coverage to adjust it to clinical scenarios and for a dose conscious approach standard dose reduction algorithm such as CARE Dose4D and CARE kV can be applied as well
- Dedicated i-Sequence mode covering full or half of the detector width which allows 3D planning and guidance

[1] *Requires a second control room monitor*

Optional system software and hardware for CT-guided interventions

myNeedle Laser[1]

Projects the needle entry point and insertion angle on to the patient with a maximum deviation of 5 mm. myNeedle Laser is embedded in the CT gantry and fully integrated in the myNeedle Guide 3D software to increase efficiency compared to external laser guidance systems.

i-Fluoro[2,3]

Allows for near real-time CT fluoroscopic image guidance. The scan mode i-Fluoro CT is completely integrated in the interventional workflow of myNeedle Guide. i-Fluoro lets you scan continuously, and view images in real time at up to 10 frames/s on an additional in-room monitor and as well on the second control room monitor. The acquired images have an image matrix of 512 x 512.

HandCARE

For i-Fluoro and i-Sequence scans HandCARE can be applied enabling real-time dose modulation to avoid direct X-ray exposure to the physician's hands. HandCARE switches off the x-ray exposure for a 100° angle between three user selectable positions (10:00, 12:00 and 2:00 o'clock).

X-Ray Foot switch

For triggering scans from the examination room

MPT Table[4]

Increased Distance
The distance between gantry and MPT Table can be increased by 20 cm with full support of myNeedle Laser and FAST 3D Camera.

i-Joystick

The i-Joystick supports the table movement in z-direction (in and out of the gantry) directly from the table side. The i-Joystick can be flexibly mounted along both sides of the table designed for an ergonomic set-up during CT-guided minimal invasive procedures. It is connected via cable and can be mounted on both sides of the CT-table.

Tablet dock for patient table

The tablet dock for the patient table is fully adjustable for an ergonomic independent in-room operation during minimal invasive procedures. Optionally the table dock can be plugged in for an uninterrupted power supply for long interventions.

[1] *Only available in combination with myNeedle Guide 3D*
[2] *Only available together with myNeedle Guide 2D or myNeedle Guide 3D*
[3] *X-Ray Foot switch required*
[4] *Optional*

Optional Radiation Therapy system software and hardware

HD FOV[1]

Designed to enable visualization of the human body parts and skin line located outside of the 50 cm scan field of view up to the bore size, based on an algorithmic complement of missing detector data outside of the 50 cm scan FOV. The image quality for the area outside the 50 cm scan field of view does not meet the image quality of the area inside the 50 cm scan field of view. Image artifacts may appear, depending on the patient setup and anatomy scanned. HD FOV cannot be used for scan FOV smaller than 50 cm.

Radiation Therapy Basic

Radiation Therapy Basic is a dedicated virtual simulation tool designed for RT available under the CT View&GO platform.Contains Sim&GO.

General features

- Concurrent display of up to a total of 2 image series (1 single or 1 fused series)
- Data pre-fetching from DICOM nodes and imaging devices, simple import from CDs and DVDs, patient data reconciliation
- Image preview function
- Drag&Drop image loading
- Automatic data transfer to TPS configuration
- DICOM, HL7 and IHE-RO standard compliance
- Create annotations and measurements and share them

Patient marking

- Single or multiple reference points or isocenters
- Absolute and relative patient marking
- Automatic marking of structure centroids
- Direct Laser Steering for compatible lasers[2]
- DICOM and text file data exchange with lasers
- Virtual Laser View for display of laser lines on 3D patient model (VRT)
- Patient marking on any supported image type
- One-click breast isocenter placement with automated contouring

Beam Placement

- Beam Placement including DRR, Source to Distance and beam templates
- Automated beam shaping based on RT structure

Contouring features

- User-configurable organ templates based on a subset of the FMA (Foundation Model of Anatomy) for interoperability between IT systems
- "CT-free" contouring: native PET or MR contouring
- Parallel contouring: contouring performed on any image is reflected on all other images
- Semi-automatic freehand contouring 2D & 3D
- Smart 2D / 3D Nudge
- Contour on any arbitrary plane including oblique planes
- Organ algebra (union, intersection, exclusion)
- Symmetric and asymmetric structure growth or contraction
- Multiple structure set support (1 per image series)
- Molecular imaging data such as PET, threshold-based and skin, gray value-based segmentation
- Visualization of previously drawn structures on the current image series

4D data management

- 4D phase splitting
- tMinIP, tMIP, AverageCT generation
- Cine-loop
- ITV generation
- Quantitative assessment of 3D tumor trajectory and amplitude and semi-automatic calculation of the midventilation phase
- Control the patient marking workflow with the dedicated RT tablet and avoid unnecessary switching between different interfaces to enter laser co-ordinates.

[1] *Part of the standard system configuration*
[2] *Available for following LAP lasers: DORADO 1, 3, 4, DORADOnova 1, 3, 5*

Optional Radiation Therapy system software and hardware

DirectORGANS

DirectORGANS (Optimized Recon based Generative Adversarial NetworkS) is the world's first auto-contouring feature directly on the scanner. It leverages the power of dedicated reconstruction and deep learning to streamline organs-at-risk contouring.

DirectORGANS Advanced[1]

DirectORGANS Advanced is an extended organ package that supports the following organs for auto-contouring.

- Lung
  - Individual Ribs
  - Sternum
  - Aorta
  - Lung Lobe LL
  - Lung Lobe UL
  - Lung Lobe LR
  - Lung Lobe MR
  - Lung Lobe UL
- Heart (iodine contrast is required)
  - Cardiac Left Ventricle
  - Cardiac Right Ventricle
  - Cardiac Right Atrium
  - Cardiac Left Atrium
- Lymphnodes
  - LN Common Iliac (L/R)
  - LN Internal Iliac (L/R)
  - LN External Iliac (L/R)
  - LN Obturator (L/R)
  - LN Presacral
- Endocardium

RadOnc Excellence

RadOnc Excellence is a dedicated RT optional package designed for the radiation oncologist and physicist.

- RT Dose Display volumes overlaid on any supported image type and side by side
- Display-related dose volume histograms
- Use deformable registration between current and prior dose volumes and images for dose accumulation
- Multimodality data handling with 3D CT, PET, PET/CT, MRI and Linac CBCT
- 4D CT, PET/CT, MRI
- Time-resolved CT and MR images
- Calculate SUV for PET images

Multimodality Deformable Registration (Contouring propagation)

- 3 different types of deformable registration
  - General deformable registration
  - Landmark-based registration
  - ROI-based registration
- Save registrations and save aligned or deformed images as a new image series
- Registration Quality Check with spyglass, deformation vector map, deformation magnitude color map
- Semi-automatic contour propagation over 4D CT breathing phases
- Quantitative 4D assessment of 3D tumor trajectory and amplitude and semi-automatic calculation of the midventilation phase

DirectDensity[2]

DirectDensity images enable kV-independent dose calculation on the treatment planning system. Personalize your scan and benefit from a patient-specific kV selection with improved soft-tissue contrast while keeping a standardized workflow without the need for additional calibration curves.

[1] *DirectORGANS is a prerequisite*
[2] *DirectDensity reconstruction is designed for use in Radiation Therapy Planning (RTP) only. DirectDensity reconstruction is not intended to be used for diagnostic imaging.*

Optional Radiation Therapy system software and hardware

RT table

Patient table 2000 mm / 307 kg RT with 307 kg Multi-index RTP Overlay
- Max. table load 307 kg / 676 lbs
- Multi-indexing with Varian and Elekta indexing
- Light-weight overlay
- TG-66 compliant

Multi-purpose Table RT with 340 kg Multi-index RTP Overlay
- Max. table load 340 kg / 750 lbs
- Multi-indexing with Varian and Elekta indexing
- Light-weight overlay
- TG-66 compliant

Respiratory Motion Management[1]

Various acquisition modes and protocols accommodate a wide range of respiratory patterns and workflows.

Extended scan time capabilities up to 200 seconds

Supports retrospective modes including phase and amplitude reconstructions

Supports the automatic creation of temporal MiniIP (tMinIP), temporal MaxIP (tMaxIP) and the easy generation via reconstruction of an Average CT, to evaluate respiratory motion

Ability to automatically detect synchronization points

User-selectable number and placement of reconstruction bins up to 1% recon

Quantitative 4D assessment of 3D tumor trajectory and amplitude and semi-automatic calculation of the midventilation phase for RT available under CT View&GO platform

FAST 4D[2]

FAST 4D streamlines your 4D CT workflow by automatically setting the optimal scan parameters based on the patient's breathing rate.

The direct online connection between the CT and a gating device over the Varian RGSC online interface or the ANZAI interface allows display and analysis of the breathing rate in real time.[2]

Gating interfaces

Varian RGSC interface
The online mode allows for retrospective gating

ANZAI Interface
The online mode allows for retrospective gating

Open Interface
For retrospective gating with a number of supported gating systems

Direct i4D[2]

Direct i4D is the world's first 4D CT sequence scan mode that adapts to the patients breathing in real time. Moreover, it intelligently adapts the reconstruction based on the breathing amplitude analysis. As a result, a reduction of 85% of image artifacts can be achieved, which would occur when not a complete inhalation and exhalation cycle was acquired[3].

Direct i4D is available for challenging cases, such as slow breathers, irregular breathing, and long scan ranges and can accommodate lung, liver, and esophagus cases.

[1] *Requires an interface to connect to one of the many compatible 3rd-party gating devices, like VARIAN RGSC or ANZAI*
[2] *Requires Varian RGSC Interface or ANZAI Interface and Respiratory Motion Management option*
[3] *Based on simulation results – Data on file*

Image quality

Low-contrast resolution

Low-contrast resolution is the ability to see ...
- a small object
- with a certain contrast difference
- on a particular phantom
- with a particular dose ($CTDI_{vol}$)

Phantom	CATPHAN (20 cm)	
Object size	5 mm	2 mm
Contrast difference	3 HU	3 HU
$CTDI_{vol}$ (32 cm)	6.0 mGy	12.0 mGy
Technique	Spiral acquisition, 10 mm slice, 80 kV, ADMIRE Strength 5	

Isotropic resolution

Isotropic 0.3 mm resolution using Siemens Healthineers' proprietary SureView technology

High-contrast resolution

x/y plane	
50% MTF	14.02 lp / cm (± 10%)
10% MTF	15.75 lp / cm (± 10%)
2% MTF	16.40 lp / cm (± 10%)
Technique	• Tungsten wire in air • 120 kV, 300 mA, 1 s, 0.6 mm, Hr72

z plane	
50% MTF	8.3 lp / cm (± 10%)
10% MTF	16.7 lp / cm (± 10%)
2% MTF	21.1 lp / cm (± 10%)
Technique	• Gold platelet phantom • 120 kV, 100 mA, 0.5 s, 0.6 mm

Homogeneity

Cross-field uniformity in a 20 cm water phantom
- max. ± 4 HU[1]
- typ. ± 2 HU

Phantom positioned near center of rotation

Dose, $CTDI_{100}$ values mGy / 100 mAs

Phantom Ø		kV	kV	kV	kV	kV
		70	80	90	100	110
16 cm	**A**	3.91	6.02	8.52	11.33	14.40
16 cm	**B**	4.36	6.54	9.09	11.96	15.09
32 cm	**A**	0.85	1.45	2.20	3.09	4.09
32 cm	**B**	2.13	3.29	4.69	6.30	8.08

Phantom Ø		kV	kV	kV	kV
		120	130	140	150
16 cm	**A**	17.73	21.24	24.95	28.92
16 cm	**B**	18.45	22.06	25.88	29.90
32 cm	**A**	5.20	6.42	7.73	9.15
32 cm	**B**	10.03	12.13	14.37	16.77

A: at center **B: 1 cm below surface**

Technique	• PMMA-Phantom – 128 x 0.6 mm • Absorbed dose for reference material air • Expected deviation: ± 10% without spectral filter ± 20% with spectral filter • Max. deviation: ± 20% without spectral filter ± 30% with spectral filter

[1] *Max. HU value also valid for 30 cm phantom*

Installation

Components

Dimensions	Height (mm / inch)	Width (mm / inch)	Length (mm / inch)	Weight (kg / lbs)
Gantry	1,995 / 78.5	2,535 / 99.8	959 / 37.8	2,180 / 4,806
CT patient table (227 kg)	≤ 950 / 37.4	≤ 700 / 27.6	≤ 2,480 / 97.6	≤ 330 / 728
CT patient table (307 kg)[1]	≤ 950 / 37.4	≤ 700 / 27.6	≤ 2,560 / 100.8	≤ 385 / 849
Multi-purpose patient table (340 kg)[1]	≤ 1,065 / 41.9	≤ 700 / 27.6	≤ 2,560 / 100.8	≤ 490 / 1,080

Power supply

Nominal voltage ± 10%
380–480 V

Nominal line frequency ± 3Hz
50, 60 Hz

Power consumption

Max. power consumption
≤ 140 kVA

Standby
≤ 3 kVA

Power consumption according to COCIR and GPP

Use scenario 24-hour power consumption[5]

Off[3]
24.8 kWh

Low power[4]
25.5 kWh

Idle (stand-by)
47.3 kWh

Protection against input power instability

Controllers
20 ms

***syngo* Acquisition Workplace**
3 min, with UPS

Frequency stability
± 5% at 50 and 60 Hz

Sound design

Standby
< 60 dB(A)

Peak
< 70 dB(A)

Electromagnetic compatibility

This product is in compliance with IEC 60601-1-2 and fulfills CISPR 11 Class A.

Room environment

Temperature range
18–30 °C / 64.4–86 °F

Relative air humidity without condensation
20–75%

Heat dissipation
(gantry, table and integrated computers)
< 12.8 kW scanning

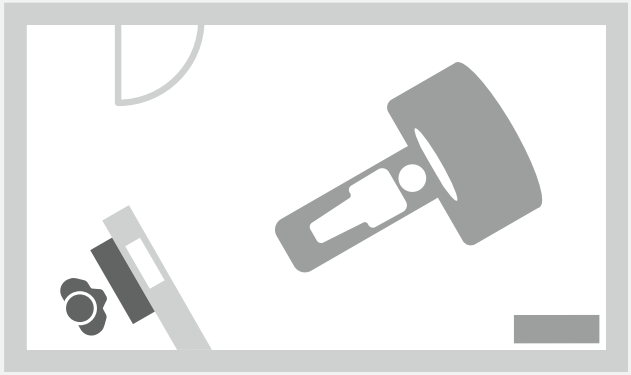
Heat dissipation (image reconstruction and acquisition system)
≤ 1.1 kW

Surface area for installation[2]

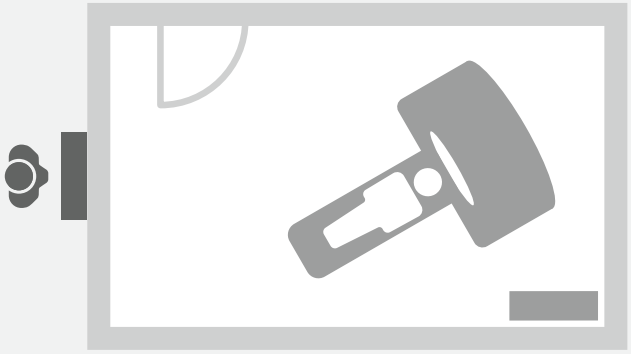
System footprint
< 17 m^2 / 183 ft^2
(14.6 m^2 / 157 ft^2 with restricted scan range)

[1] *Optional*
[2] *Subject to local regulations. Safety distances must be checked according to country-specific requirements.*
[3] *With wall-switch*
[4] *System off*
[5] *Values may vary approx. ± 3% due to specific system conditions, for example of UPS, etc.*

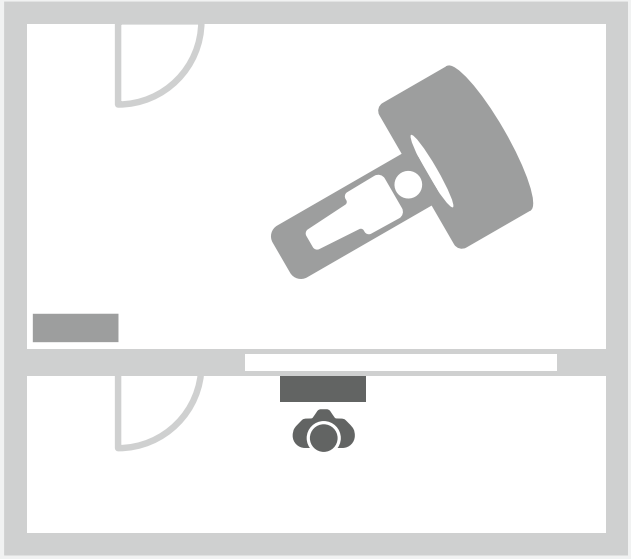
Installation



Follow the “niche” concept to work in the examination room.



Position the workstation outside the room, e.g., in the corridor.



Minimize the room requirements of a separate control room.

Notes

Notes

Notes

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